

Do Market Regimes Improve Machine-Learning Stock Ranking?

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Abstract

Can market regimes improve machine-learning stock selection and portfolio optimization? Using Bloomberg data for 206 large U.S. stocks from 2005 to 2025, we train an XGBoost model to rank stocks by predicted 20-day cross-sectional return percentiles. A monthly equal-weight portfolio of the ten highest-ranked stocks earns a 34.24% CAGR, a Sharpe ratio of 1.279, and a SPY-adjusted single-factor alpha of 13.55% during the 2021–2025 out-of-sample period. The strongest corrected statistical evidence is for outperformance relative to SPY and random Top10 portfolios. The portfolio significantly outperforms SPY, random Top10 portfolios, and predicted low-ranked stocks. However, adding Hidden Markov Model regimes as features, training regime-specific models, and using regime-dependent portfolio optimization do not improve performance. Regimes help explain when the strategy works, but the stock-ranking model itself is the main source of value.

Keywords: Machine learning; stock ranking; XGBoost; portfolio optimization; market regimes; Hidden Markov Models; cross-asset signals; transaction costs.

1 Introduction

Stock selection is fundamentally a ranking problem. Investors rarely need to forecast the exact return of every stock. They need to identify which stocks are more likely to outperform the cross-section and then decide how to allocate capital to those stocks. Machine learning is well suited to this problem because it can combine nonlinear information from prices, fundamentals, volatility, momentum, and cross-asset markets.

This paper asks whether market regimes improve that ranking problem. The question matters because return-predictor relations are unlikely to be constant across market environments. A momentum signal, a volatility signal, or a valuation signal may behave differently in a calm bull market than in a correction or crisis. If such regime dependence is economically important, then incorporating regime information should improve either the ranking model or the portfolio construction step.

We test this idea using Bloomberg data for approximately 206 large U.S. stocks from 2005 to 2025. The forecasting model is an XGBoost regressor trained to predict future 20-day cross-sectional return rank percentiles. Each month, stocks are ranked by their predicted score, and the main portfolio invests equally in the ten highest-ranked stocks. The model is trained on 2005–2018 data, validated on 2019–2020 data, and evaluated out of sample from 2021 to 2025.

The baseline model performs strongly. The equal-weight Top10 portfolio earns a 34.24% CAGR, compared with 14.72% for SPY. It also has a Sharpe ratio of 1.279, compared with 0.746 for SPY, and a maximum drawdown of -12.27%, compared with

-20.47% for SPY. The Top10 portfolio significantly outperforms SPY, random Top10 portfolios, and predicted low-ranked stocks. The ranking evidence is also positive: the average monthly Spearman information coefficient is 0.0424 and is statistically significant.

The main result is not that every extension improves the model. It is the opposite. The simple ranking portfolio is difficult to improve. Alternative machine learning models such as Random Forest, Extra Trees, LightGBM, Neural Networks and Ridge regression do not outperform XGBoost overall. Removing the largest mega-cap technology and growth stocks reduces returns but preserves the main statistical evidence, suggesting that the strategy is not driven only by a small set of dominant winners.

The regime results are especially informative. We estimate an expanding-window Gaussian Hidden Markov Model using only information available up to each date. The HMM identifies economically meaningful market states, and the baseline strategy performs differently across them. However, using the HMM regimes directly does not improve out-of-sample performance. Adding regime indicators as features reduces the Top10 CAGR from 34.24% to 30.37%. Training separate XGBoost models within each regime performs worse, with a CAGR of 23.44%. These results suggest that regimes contain descriptive information about market conditions, but they do not provide enough incremental forecasting value to improve the stock-ranking model.

Portfolio optimization also fails to improve the main result. Minimum-variance, score-aware mean-variance, HMM covariance, and HMM score-aware mean-variance portfolios are all tested inside the ML-selected Top10 universe. None outperforms the simple equal-weight Top10 strategy. This finding is consistent with a practical interpretation: once the model has identified a concentrated set of attractive stocks, imposing additional covariance-based optimization adds estimation error and concentration without improving performance.

The paper contributes to the empirical asset-pricing and machine-learning literature in three ways. First, it shows that a cross-sectional rank-percentile target can produce economically meaningful and statistically significant stock-selection performance. Second, it separates prediction from interpretation: HMM regimes help explain when the model performs well, but they do not improve the forecasts themselves. Third, it shows that simple equal weighting can dominate more complex portfolio optimization when expected-return estimates come from noisy machine-learning scores.

The main takeaway is straightforward. In this setting, the value comes from ranking stocks well, not from making the model more regime-aware or the portfolio optimizer more sophisticated. Market regimes matter for understanding conditional performance, but the strongest implementable strategy remains simple: each month, buy the ten stocks with the highest predicted cross-sectional return ranks and weight them equally.

2 Literature Review

2.1 Regime-Switching Models in Economics and Finance

The modern econometric literature on regimes begins with [Hamilton \(1989\)](#), who introduces a Markov-switching framework for modelling discrete shifts in macroeconomic time series. Hamilton’s contribution was important because it formalized the idea that economic and financial variables may follow different processes in different unobserved states, rather than being generated by a single stationary model. In the context of asset returns, this matters because expected returns, volatility, and cross-asset correlations may differ between expansions, contractions, recoveries, and crises. Hamilton’s framework therefore provides the econometric foundation for the regime-detection component of this paper.

The relevance of regime-switching models for portfolio allocation was developed more directly by Ang and Bekaert. In their study of international asset allocation with regime shifts, [Ang and Bekaert \(2002\)](#) show that correlations and volatilities tend to rise in highly volatile bear markets, reducing the benefits of diversification exactly when investors need them most. Their work is directly relevant to this paper because it motivates the possibility that both forecasting relationships and portfolio risk estimates should be conditioned on the prevailing market state. [Ang and Bekaert \(2004\)](#) further connect regimes to practical asset allocation by emphasizing that asset allocation decisions can change materially when return distributions are state dependent.

[Guidolin and Timmermann \(2007\)](#) extend this literature by studying asset allocation under multivariate regime switching. They find evidence for multiple regimes, including crash, slow-growth, bull, and recovery states, and show that optimal asset allocations vary substantially across these states. Their result is central to the motivation of this paper: if regimes change both return distributions and risk characteristics, then regime information may be useful not only for forecasting but also for portfolio construction. Relatedly, [Perez-Quiros and Timmermann \(2000\)](#) show that firm-size return dynamics vary over the

business cycle, supporting the broader idea that cross-sectional equity relationships may be state dependent. This is especially relevant for a stock-ranking model because the predictive value of characteristics such as size, momentum, valuation, and volatility may vary across regimes.

2.2 Hidden Markov Models and Regime-Based Portfolio Allocation

Hidden Markov Models are a natural empirical tool for regime detection because market states are not directly observed. Instead, they are inferred from observable return, volatility, drawdown, and macro-financial variables. The statistical foundations of HMMs are treated extensively in [Cappé et al. \(2005\)](#), [Frühwirth-Schnatter \(2006\)](#), and [Kim and Nelson \(1999\)](#). These references justify the use of latent-state models when the researcher believes that observed time series are generated by different underlying regimes.

Several empirical studies apply HMMs directly to finance. [Bulla and Bulla \(2006\)](#) show that HMMs and hidden semi-Markov models can reproduce many stylized facts of financial return series, although standard HMMs may struggle with duration dependence and volatility persistence. This is relevant to the present study because it supports the use of HMMs for identifying market states, while also warning that regime models are imperfect approximations of financial dynamics. [Hassan and Nath \(2005\)](#) represent an early application of HMMs to stock market forecasting. Their work is less directly related to modern cross-sectional asset pricing, but it is useful historically because it shows the long-standing interest in HMMs for financial prediction.

The closest portfolio-allocation literature links HMMs to dynamic asset allocation. [Bae et al. \(2014\)](#) use HMMs to identify regimes and incorporate regime-dependent distributions into a stochastic programming framework. Their study is particularly relevant because it treats regimes as a way to transform static mean-variance allocation into an optimization problem under uncertainty. [Nystrup et al. \(2018\)](#) similarly study dynamic portfolio optimization across hidden market regimes and show how HMM-based state information can guide portfolio decisions. [Erlwein-Sayer et al. \(2020\)](#) study filter-based portfolio strategies in an HMM setting with varying correlation parametrizations, which is relevant to this paper’s HMM covariance-based portfolio optimization experiments.

This paper differs from much of the HMM allocation literature in an important way. Rather than applying regime detection to a small set of asset classes, it first uses machine learning to rank individual stocks and then tests whether regime information improves either the ranking model or the portfolio construction step. This distinction matters because regime-aware allocation across asset classes may be easier than regime-aware stock selection across hundreds of securities. In a large cross-section, stock-level noise, model error, and covariance estimation error may dominate the benefits of regime conditioning.

A more recent literature applies regime-switching models to factor investing and large-scale equity portfolios. [Wang et al. \(2020\)](#) use an HMM to identify U.S. stock market regimes and propose switching among factor strategies depending on the detected regime. Their paper is especially relevant because it connects HMMs with factor timing rather than only broad asset allocation. The present study is related but distinct: instead of switching among pre-defined factor portfolios, it trains a machine learning model to rank individual stocks using a broad set of technical, fundamental, momentum, and cross-asset variables. Recent work by [Ibanez and Urga \(2024\)](#) is even closer to the large-scale stock portfolio setting. They propose a Hidden Markov Model approach for incorporating market regimes into large-scale stock portfolios through regime-conditional factors. Their study is relevant because it asks whether regime-dependent behavior can improve systematic equity portfolio construction. This paper contributes to the same question from a different angle: it tests whether regime information improves machine-learning-based stock rankings and whether regime-specific covariance estimates improve Top10 portfolio construction.

The recent comparison by [Baitinger and Hoch \(2024\)](#) is also relevant because it highlights the danger of unnecessary regime-model complexity. Their results suggest that although hidden semi-Markov models may improve in-sample fit, their out-of-sample advantage over simpler HMMs may largely disappear. This finding supports the conservative design choice in the present paper: a four-state Gaussian HMM is used as an interpretable regime classifier, but the paper does not assume that a more complex regime model will necessarily improve investment performance. This is consistent with the empirical finding that HMM regimes help explain conditional performance but do not improve aggregate forecasting or portfolio construction.

2.3 Machine Learning in Empirical Asset Pricing

The machine learning component of this paper is motivated by the modern empirical asset pricing literature. [Gu et al. \(2020\)](#) provide one of the most influential benchmarks, showing that machine learning methods can generate large economic gains in expected-return prediction and that tree-based methods and neural networks benefit from modelling nonlinear predictor interactions. This directly supports the use of XGBoost in the present paper, since the model can capture nonlinear interactions among momentum, volatility, fundamentals, market conditions, and cross-asset variables.

[Kelly et al. \(2019\)](#) propose Instrumented Principal Component Analysis, linking firm characteristics to latent factor exposures and time-varying loadings. [Kozak et al. \(2020\)](#) study high-dimensional cross-sectional predictors and emphasize the importance of shrinkage and robustness in expected-return modelling. Together, these papers justify the use of many predictors while also warning against overfitting.

The present paper differs from much of the expected-return prediction literature by formulating the target as a future cross-sectional return rank percentile rather than raw future returns. This design is closely aligned with stock selection. A portfolio that buys the Top10 stocks does not require calibrated return forecasts; it requires a model that tends to assign higher scores to future winners than to future losers. Therefore, ranking metrics such as the Spearman information coefficient and Top-minus-Bottom returns are more relevant than traditional R^2 alone.

2.4 Portfolio Construction Under Estimation Error

The portfolio construction tests in this paper are grounded in the classic mean-variance tradition introduced by [Markowitz \(1952\)](#), where investors choose portfolios by balancing expected return and variance. In principle, once a machine learning model provides expected-return information, mean-variance optimization should be able to improve portfolio performance by using both expected-return and covariance estimates.

However, a large literature shows that optimized portfolios are highly sensitive to estimation error. [Ledoit and Wolf \(2004\)](#) emphasize that the sample covariance matrix contains estimation error that is particularly damaging for portfolio optimization and propose shrinkage as a remedy. [DeMiguel et al. \(2009\)](#) show that the naive $1/N$ allocation is difficult to outperform out of sample, because optimized portfolios often require very accurate estimates of expected returns and covariances. This literature is essential for interpreting the results of this paper: the equal-weight Top10 portfolio outperforms more sophisticated static and HMM-based optimization methods, suggesting that the additional estimation error introduced by optimization outweighs its theoretical benefits.

The portfolio optimization experiments are therefore not a secondary detail but a direct test of an important practical question. If machine learning identifies a strong Top10 set, should the investor then optimize weights? The empirical answer in this study is no. Equal weighting remains superior because it preserves the ranking signal while avoiding unnecessary sensitivity to noisy covariance and expected-score estimates.

2.5 Positioning and Research Questions

The existing literature suggests that regimes matter for asset returns, that HMMs can identify latent market states, and that machine learning can improve cross-sectional return prediction. However, less is known about whether HMM-based regime information improves machine-learning stock ranking and portfolio construction in a large individual-equity universe. This paper addresses that gap.

The paper contributes by combining three elements that are often studied separately: machine-learning-based stock ranking, expanding-window HMM regime detection, and portfolio construction within the ML-selected stock set. The empirical results refine the regime-switching hypothesis. Regimes are economically meaningful and help explain when the strategy performs well or poorly, but using regimes directly as forecasting inputs or as the basis for separate models does not improve out-of-sample performance. Similarly, HMM-based portfolio optimization does not outperform equal weighting. The contribution is therefore not that regimes always improve investment strategies, but that their value depends on where they enter the investment process. In this study, regimes are more useful as diagnostic and interpretive tools than as direct sources of forecasting or portfolio-construction improvement.

2.5.1 Research Question

The main research question is:

Can market regime information improve machine-learning-based stock ranking and portfolio construction?

This question is decomposed into three empirical subquestions:

- Stock-ranking question: *Can a machine learning model trained on technical, fundamental, momentum, and cross-asset variables generate statistically significant cross-sectional stock rankings?*
- Regime-forecasting question: *Does adding HMM-based market regime information improve the predictive quality of the stock-ranking model?*
- Regime-portfolio question: *Does using regime information in portfolio optimization improve realized portfolio performance relative to a simple equal-weight Top10 strategy?*

The final results suggest a nuanced answer. Machine learning stock ranking is effective in this dataset, and market regimes explain variation in strategy performance across time. However, explicitly incorporating regime information into the forecasting model or portfolio optimizer does not improve aggregate out-of-sample performance.

2.5.2 Contributions

This paper makes five contributions.

First, it studies stock selection as a cross-sectional ranking problem rather than as a direct return forecasting problem. The final target is the future rank percentile of each stock’s 20-trading-day return within the investment universe on each date. This is better aligned with portfolio construction than raw return prediction because the investment rule depends on relative ranking rather than calibrated expected returns.

Second, the paper develops a Bloomberg-based empirical pipeline using adjusted OHLCV data, fundamentals, technical indicators, momentum features, and cross-asset market variables. The final baseline uses 68 features and is tested out of sample over 60 monthly rebalances from 2021 to 2025.

Third, the paper compares several feature sets and shows that cross-asset market variables improve performance, whereas slower-moving macroeconomic variables and interest-rate variables degrade performance. This distinction is economically meaningful: traded cross-asset variables appear to provide useful market-state information, while infrequently updated macro variables add noise in this cross-sectional ranking setting.

Fourth, the paper evaluates market regimes using an expanding-window Hidden Markov Model. The HMM avoids full-sample look-ahead bias by training only on data available up to the current date and refitting monthly. The experiments show that regime information is informative for conditional performance analysis but does not improve the global forecasting model or the portfolio optimization stage.

Fifth, the paper evaluates realistic portfolio construction and implementation issues. It shows that equal weighting outperforms several optimized portfolios, consistent with the idea that estimation error in covariance and expected-return inputs can dominate theoretical optimization benefits. The strategy remains statistically significant relative to SPY up to 25 basis points before multiple-testing correction. After correction, the strongest evidence remains at lower and moderate cost assumptions, while the 25-basis-point result should be interpreted as economically robust but statistically more marginal.

3 Data and Methodology

3.1 Equity Universe and Data

3.1.1 Equity Universe

The equity universe consists of approximately 206 U.S. stocks. The list contains large, liquid U.S. equities across sectors, including major technology, financial, healthcare, consumer, industrial, energy, utility, and real estate names.

3.1.2 Data Extraction

The model uses Bloomberg daily OHLCV data from 2005 to 2025. SPY is adjusted using total-return information to better match dividend-inclusive benchmark performance.

Bloomberg fundamentals are included when available: market capitalization, price-to-earnings ratio, price-to-book ratio, return on equity, debt-to-equity ratio. Market capitalization is log-transformed. Fundamental variables are merged through backward-looking date alignment and forward-filled within ticker. Missing values are filled cross-sectionally by date using the median when necessary.

The final baseline includes market and cross-asset variables: SPY returns and volatility, VIX level and changes, U.S. aggregate bond return over 20 and 60 days, U.S. dollar index return over 20 and 60 days, gold return over 20 and 60 days, oil return over 20 and 60 days. Interest-rate and macro-cycle variables were tested but excluded from the final baseline because they degraded out-of-sample performance.

3.1.3 Sample Split

The empirical design uses a chronological split: The training period runs from 2005-01-01 to 2018-12-31, the validation period from 2019-01-01 to 2020-12-31, and the test period from 2021-01-01 to 2025-12-31.

The final baseline contains: 654,009 training rows; 102,527 validation rows; 253,665 test rows; 68 features; 206 stocks.

Notation

Let $\mathcal{U}_t$ denote the set of stocks available on date t , with cardinality $N_t = |\mathcal{U}_t|$. For stock $i \in \mathcal{U}_t$, let $P_{i,t}$ denote the adjusted closing price and let H denote the prediction horizon. In the baseline specification, $H = 20$ trading days. The forward simple H -day return is defined as

$$R_{i,t}^{(H)} = \frac{P_{i,t+H}}{P_{i,t}} - 1. \quad (1)$$

The vector of observable predictors for stock i at date t is denoted by

$$X_{i,t} = \left(x_{i,t}^{(1)}, x_{i,t}^{(2)}, \dots, x_{i,t}^{(K)} \right), \quad (2)$$

where K is the number of features used by the model. In the final baseline specification, $K = 68$.

3.2 Feature Engineering

Feature engineering is designed to capture stock-level momentum, relative strength, volatility, technical patterns, fundamental characteristics, and cross-asset market conditions. All features are constructed using information available at or before the prediction date. The main text summarizes the economic motivation for each feature group, while detailed definitions of standard technical indicators are reported in [Appendix A](#).

3.2.1 Return and Momentum Features

The model includes stock returns over several windows: 1, 2, 5, 10, 20, 60, 120, and 252 trading days. Longer-horizon momentum variables include: 3-month momentum; 6-month momentum; 12-month momentum; with 12-month momentum skipping the most recent month. For a lookback window h , the backward-looking return used as a predictor is

$$R_{i,t}^{(-h)} = \frac{P_{i,t}}{P_{i,t-h}} - 1. \quad (3)$$

The model includes several such windows, $h \in \{1, 2, 5, 10, 20, 60, 120, 252\}$, allowing it to capture short-term reversal, medium-term trend, and long-term momentum effects.

The skip-month momentum variable is included because short-term reversal can contaminate raw 12-month momentum. The skip-month momentum variable is defined as

$$MOM_{i,t}^{12m,skip} = \frac{P_{i,t-20}}{P_{i,t-252}} - 1. \quad (4)$$

This construction excludes the most recent month in order to reduce contamination from short-term reversal effects.

3.2.2 Relative Strength

Relative strength is measured by subtracting SPY returns from individual stock returns over matching horizons. This captures whether a stock has outperformed or underperformed the market over the same lookback window.

Let $R_{m,t}^{(-h)}$ denote the corresponding backward-looking return of the market benchmark, proxied by SPY. Relative strength is computed as

$$RS_{i,t}^{(h)} = R_{i,t}^{(-h)} - R_{m,t}^{(-h)}. \quad (5)$$

A positive value indicates that stock i has outperformed the market over the lookback window h .

3.2.3 Trend, Volatility, and Technical Indicators

The feature set also includes standard technical indicators designed to capture trend strength, short-term reversal, volatility, and trading activity. Trend variables include price relative to 20-day, 50-day, and 200-day moving averages, the ratio of the 20-day to 50-day moving average, and MACD-based indicators. These variables summarize whether a stock is trading above or below recent trend levels and whether short-term price trends are strengthening or weakening.

Volatility and risk variables include realized volatility over 5, 10, 20, and 60 trading days, average true range as a percentage of price, rolling beta to SPY over 20 and 60 trading days, and rolling correlation with SPY over the same horizons. These variables allow the model to distinguish between stable winners, highly volatile stocks, and stocks whose returns are closely linked to the market.

Finally, short-term oscillator and volume variables include RSI, stochastic %K, stochastic %D, Williams %R, one-day volume change, and volume relative to its 20-day moving average. These indicators capture whether a stock is close to the upper or lower end of its recent trading range and whether recent price movements are accompanied by abnormal trading activity. Detailed definitions of the technical indicators are provided in [Appendix A](#).

3.3 Fundamental variables

The final model includes log market capitalization, P/E, P/B, ROE, and debt-to-equity. These variables capture size, valuation, profitability, and leverage characteristics.

3.3.1 Cross-Asset Variables

The final baseline includes returns on aggregate bonds, the U.S. dollar index, gold, and oil over 20-day and 60-day windows. These variables are included because cross-asset markets reflect changes in inflation expectations, liquidity, global risk appetite, and macro-financial conditions in real time. The feature-ablation results show that these cross-asset variables improve stock-ranking performance.

3.4 Market Regime Detection

A market regime is a persistent state of the market environment, often associated with different levels of volatility, trend, drawdown, liquidity, risk appetite, or macro-financial conditions. Examples include calm expansion, volatile correction, crisis, and recovery regimes. Regime-aware investing is motivated by the possibility that asset return dynamics and predictor relevance vary across regimes.

In a machine-learning stock-ranking setting, regime information can enter the process in several ways. One approach is to include the current regime as a feature in the model. This allows the machine learning algorithm to learn interactions between regime state and other predictors. For example, momentum may be more effective in trending regimes, while valuation or low-volatility signals may matter more during stressed markets.

A second approach is to train one model per regime. This assumes that the mapping from predictors to future returns is sufficiently different across regimes to justify specialized models. However, this approach has a cost: each model receives only the subset of observations belonging to its regime, which increases estimation variance and overfitting risk.

A third approach is to keep the forecasting model unchanged and use regimes only in portfolio construction. For example, covariance matrices may be estimated using only historical observations from the same regime, or risk aversion may be adjusted by regime. This can be useful if the primary benefit of regime detection lies in risk estimation rather than return prediction.

This paper tests all three ideas. The results indicate that regimes are useful for interpretation but not for improving aggregate performance.

A key component of the empirical framework is the identification of market regimes using a Hidden Markov Model (HMM). The motivation for using an HMM is that financial markets are not stationary: the statistical behavior of returns, volatility, drawdowns, and risk sentiment changes substantially across time. Periods of expansion, stress, crisis, and recovery are likely to be governed by different return-generating processes. A regime-switching model therefore provides a way to classify the market environment and to incorporate this information into subsequent forecasting and portfolio-construction stages.

A Hidden Markov Model assumes that the observed market variables are generated by an unobserved latent state process. Let $Z_t \in \{1, \dots, K\}$ denote the unobserved market regime at date t . The latent state process is assumed to follow a first-order Markov chain:

$$\Pr(Z_t = k \mid Z_{t-1} = j, Z_{t-2}, \dots) = \Pr(Z_t = k \mid Z_{t-1} = j) = a_{jk}, \quad (6)$$

where a_{jk} is the transition probability from regime j to regime k .

Let M_t denote the vector of observed market-level regime features. Conditional on the hidden state, the Gaussian HMM assumes

$$M_t \mid (Z_t = k) \sim \mathcal{N}(\mu_k, \Sigma_k), \quad (7)$$

where μ_k and Σ_k are the state-specific mean vector and covariance matrix.

In this context, the latent state represents the prevailing market regime. The regime itself is not directly observable, but it can be inferred from observable market indicators. The HMM estimates both the probability distribution of the observed variables conditional on each hidden state and the transition probabilities governing movements between regimes.

The HMM parameters are

$$\Theta = \left(\pi, A, \{\mu_k, \Sigma_k\}_{k=1}^K \right), \quad (8)$$

where π is the initial state distribution, $A = (a_{jk})$ is the transition matrix, and (μ_k, Σ_k) are the Gaussian emission parameters.

The likelihood of the observed market sequence $M_{1:T}$ is obtained by summing over all possible hidden state paths:

$$\mathcal{L}(\Theta; M_{1:T}) = \sum_{Z_1=1}^K \cdots \sum_{Z_T=1}^K \Pr(Z_1) \prod_{t=2}^T \Pr(Z_t \mid Z_{t-1}) \prod_{t=1}^T p(M_t \mid Z_t; \Theta). \quad (9)$$

In practice, this likelihood is evaluated recursively using the forward algorithm.

The regime model was estimated using a Gaussian Hidden Markov Model with four latent states. The input variables were selected to capture different dimensions of the market environment:

- S&P 500 log return;
- 20-day realized volatility;
- 60-day market momentum;
- 252-day drawdown;
- VIX level;
- 10-year minus 2-year Treasury yield spread.

These variables jointly represent return dynamics, risk conditions, market stress, and macro-financial conditions. All variables were standardized before model estimation to prevent differences in scale from dominating the regime classification.

Several HMM specifications were examined before selecting the final model. Models with a number of hidden states ranging from 1 to 7 were compared using information criteria, estimation stability, and economic interpretability. As shown in Figure 1, both AIC and BIC decline as the number of regimes increases, indicating that more flexible models achieve a better in-sample likelihood. However, the marginal improvement in BIC decreases substantially after four regimes. This suggests that additional states continue to improve statistical fit, but with diminishing incremental gains.

For this reason, the final specification does not select the number of regimes purely by minimizing the information criteria. Instead, the four-state model is chosen as a parsimonious compromise between goodness of fit, estimation stability, and economic interpretability. In particular, four states allow the model to distinguish between normal expansion, bull market, correction or bear-market conditions, and crisis or extreme-stress periods. A three-state model tends to merge some

economically distinct environments, especially normal and stressed conditions, while models with more than four states provide additional statistical refinement but weaker interpretability for the purposes of the subsequent stock-ranking and portfolio-construction experiments.

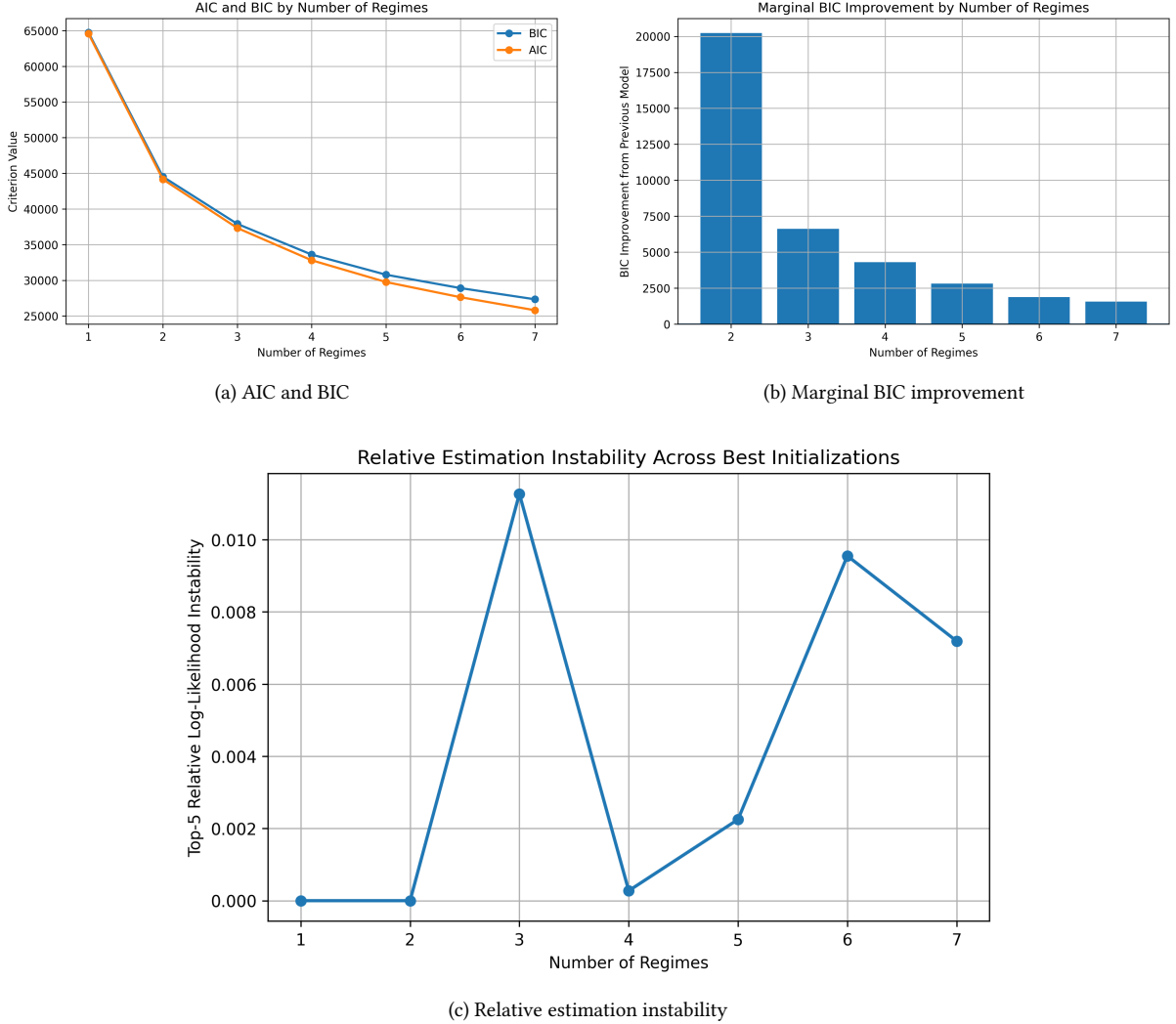


Figure 1: HMM model-selection diagnostics. AIC and BIC improve as additional regimes are introduced, but the marginal BIC gain declines after four regimes. The instability diagnostic provides an additional check on whether the likelihood surface is stable across alternative random initializations. The four-state model is selected as a parsimonious compromise between fit, stability, and economic interpretability.

Figure 1 supports the use of four regimes. The largest improvement occurs when moving from one to two states, which confirms that a single-regime model is too restrictive for market data. Additional regimes continue to improve the likelihood, but the incremental gain becomes progressively smaller. The four-state model lies at the point where the model has enough flexibility to separate economically meaningful market environments while avoiding unnecessary fragmentation of the state space. The instability diagnostic also supports this choice, since the four-state model is substantially more stable than some neighbouring specifications while remaining interpretable.

Alternative specifications were also considered. A Gaussian mixture HMM was tested to allow for more flexible within-regime distributions, but it produced unstable and overly complex results, including covariance-related warnings and less interpretable states. Conditional on the hidden state, the Gaussian mixture HMM assumes

$$M_t \mid (Z_t = k) \sim \sum_{j=1}^{J_k} \pi_{kj} \mathcal{N}(\mu_{kj}, \Sigma_{kj}), \quad (10)$$

where J_k is the number of Gaussian mixture components associated with hidden state k , π_{kj} are the mixture weights

satisfying $\sum_{j=1}^{J_k} \pi_{kj} = 1$, and μ_{kj} and Σ_{kj} denote the mean vector and covariance matrix of the j th Gaussian component within state k .

Different covariance structures were also considered. A diagonal covariance structure is more parsimonious but assumes that the regime features are conditionally independent within each state, which is a strong assumption given the natural relationship between volatility, drawdowns, returns, and VIX. The final specification therefore retained a full covariance Gaussian HMM while using a reduced and economically motivated feature set to improve numerical stability.

An important methodological issue was avoiding look-ahead bias. To address this issue, an expanding-window HMM procedure was implemented. At each forecasting date, the HMM was re-estimated using only information available up to that date. Features were standardized using a scaler fit only on the available training window. The expanding HMM procedure begins prediction on 2008-01-01 after at least 500 training observations and refits approximately every 20 trading days. At each date, the HMM is trained using observations available up to that date only.

This design avoids full-sample look-ahead bias. The regime assigned to date t is based only on information available up to t . For implementation, this is consistent with trading after the close of date t . If a same-day open execution assumption were required, the regime signal should be lagged by one trading day.

To avoid look-ahead bias, the regime estimate at date t is based only on the information set

$$\mathcal{F}_t = \sigma(M_1, \dots, M_t). \quad (11)$$

The filtered probability of regime k is therefore

$$\gamma_t(k) = \Pr(Z_t = k \mid \mathcal{F}_t). \quad (12)$$

The predicted regime used in the empirical tests is

$$\hat{Z}_t = \arg \max_{k \in \{1, \dots, K\}} \gamma_t(k). \quad (13)$$

After estimation, the numerical HMM states were mapped to economically meaningful labels. Because HMM state labels are arbitrary, the states were classified according to their estimated characteristics. The crisis state was identified as the regime with the highest stress score, combining high VIX, high realized volatility, and large drawdowns. The bull state was identified as the calmest and strongest-return state, characterized by relatively high returns, low volatility, low VIX, and limited drawdowns. Among the remaining states, the more stressed state was labelled bear, while the final state was labelled normal. This labelling procedure ensured that regime names were assigned based on economic interpretation rather than arbitrary state numbers.

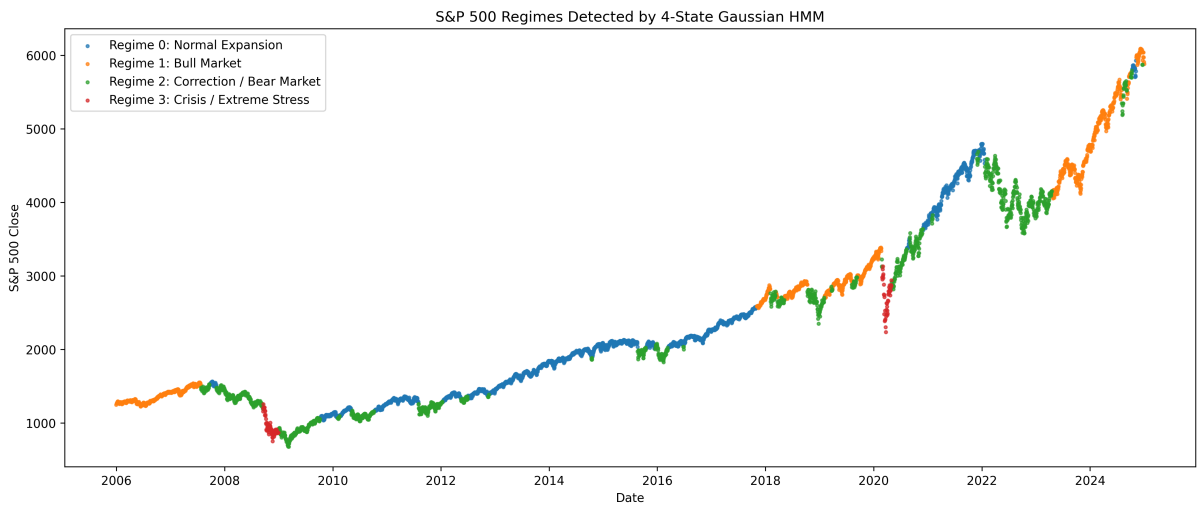


Figure 2: S&P 500 regimes detected by the four-state Gaussian HMM. The figure plots the S&P 500 level over time, with observations colored according to the estimated HMM regime. Regime labels are assigned ex post according to the economic characteristics of each state: normal expansion, bull market, correction or bear-market conditions, and crisis or extreme stress.

Figure 2 provides an economic interpretation of the four-state HMM classification. The model identifies prolonged normal-expansion periods, strong bull-market phases, correction or bear-market intervals, and crisis-like episodes. The crisis or extreme-stress regime is concentrated around the most severe market drawdowns (2008 and COVID-19 crises), while the correction or bear-market regime captures less extreme but still adverse market environments. The bull-market regime appears during strong upward-trending periods, whereas the normal-expansion regime captures more moderate market growth. This visual evidence supports the interpretation that the HMM states are not arbitrary statistical clusters, but correspond to economically meaningful market environments.

The expanding-window regime classifications were compared against full-sample HMM classifications as a diagnostic exercise. The agreement rate improved substantially relative to the earlier fixed-HMM approach, reaching approximately 65.6% during the out-of-sample period. The strongest agreement occurred for the most economically important regimes. Observations classified as bull markets by the expanding-window HMM were consistently classified as bull markets by the full-sample HMM, and most observations classified as bear markets were also classified as bear markets ex post. Disagreement was concentrated mainly in the normal regime, which is expected because normal conditions often represent transitional periods between clearer bull and bear environments.

The full-sample HMM did not assign any observations in the 2022–2025 test period to the crisis regime. This is economically plausible because the most extreme crisis episodes in the full sample are the 2008 Global Financial Crisis and the 2020 COVID crash. By contrast, the expanding-window HMM occasionally classified severe but less extreme periods in the test sample as crisis-like. These cases were typically classified as bear periods by the full-sample model, indicating that the expanding-window model was identifying heightened stress conditions, even if they were not as extreme as the largest crises in the full dataset.

Overall, the final HMM specification uses four economically interpretable regimes, a parsimonious set of market and macro-financial features, full covariance Gaussian emissions, and an expanding-window estimation procedure that avoids look-ahead bias. The regime-detection framework is therefore suitable for use as an input into the subsequent machine-learning forecasting and portfolio-construction stages.

3.5 Machine Learning Ranking Model

3.5.1 Target Variable

The final target is the future cross-sectional rank percentile of each stock’s 20-trading-day return. For each date t , stocks are ranked by their realized future return over the next $H = 20$ trading days. The target variable is the future return rank percentile:

$$y_{i,t} = \frac{\text{rank}_{j \in \mathcal{U}_t} \left(R_{i,t}^{(H)} \right)}{N_t}, \quad y_{i,t} \in (0, 1]. \quad (14)$$

A value close to one indicates that stock i was among the best future performers in the cross-section on date t , whereas a value close to zero indicates that it was among the worst future performers. Ranks are computed in ascending order of realized future return, so the highest future return receives the highest rank percentile, where $(R_{i,t,t+20})$ is the forward 20-trading-day return of stock i , and the rank percentile is computed cross-sectionally across all stocks available on date t . This target is chosen because the investment strategy depends on relative ranking rather than calibrated expected returns. It is therefore more appropriate than raw return regression for stock selection.

3.5.2 Baseline Model

The final baseline is an XGBRegressor with the following parameters:

- 600 estimators
- maximum depth 4
- learning rate 0.03;
- subsample 0.8;
- column sample by tree 0.8;
- minimum child weight 10;
- L2 regularization 2.0;

- regression objective: squared-error

Although the investment problem is ultimately a ranking problem, the model is estimated as a regression on the rank-percentile target. For a training sample $\mathcal{T}$, the model learns a function $f(\cdot)$ by minimizing squared error:

$$\hat{f} = \arg \min_{f \in \mathcal{F}} \sum_{(i,t) \in \mathcal{T}} (y_{i,t} - f(X_{i,t}))^2 + \Omega(f), \quad (15)$$

where $y_{i,t}$ is the realized future rank percentile, $X_{i,t}$ is the feature vector, and $\Omega(f)$ represents the regularization terms used by gradient-boosted trees.

The model output is therefore

$$\hat{s}_{i,t} = \hat{f}(X_{i,t}), \quad (16)$$

which is interpreted as a predicted future rank score rather than a calibrated expected return.

The model is trained as an ensemble over seven random seeds (1, 10, 42, 123, 500, 999, and 2025). Predictions are averaged across seeds. This reduces sensitivity to stochastic variation in tree construction. Let $\hat{f}^{(b)}$ denote the model trained with random seed b , for $b = 1, \dots, B$. The ensemble score is computed as

$$\hat{s}_{i,t} = \frac{1}{B} \sum_{b=1}^B \hat{f}^{(b)}(X_{i,t}), \quad (17)$$

where $B = 7$ in the final specification.

3.5.3 Alternative Models and Robustness Protocol

The following alternative models are evaluated under the same features, target, train-validation-test split, portfolio construction, and evaluation metrics: Random Forest Regressor; Extra Trees Regressor; Ridge Regression; LightGBM Regressor; Neural Network (MLPRegressor). These models are not extensively hyperparameter-optimized. Instead, standard conservative settings are used to compare model classes under a common protocol. The objective is not to find the best possible version of every model, but to determine whether XGBoost is a reasonable and competitive choice.

Randomized hyperparameter searches and an early-stopping specification were also evaluated as robustness checks on the baseline model configuration; results for the alternative models, the hyperparameter search, and the early-stopping check are reported together in Section 5.2.

3.6 Portfolio Construction

3.6.1 Baseline Portfolio

At each monthly rebalance date, the model predicts scores for all available stocks. Stocks are sorted by predicted score, and the portfolio invests equally in the Top 10 names. Each stock receives a 10% weight. The holding horizon is 20 trading days, approximately one trading month.

At each rebalance date t , stocks are sorted by their predicted score $\hat{s}_{i,t}$. The selected Top10 set is

$$\mathcal{P}_t^{Top10} = \{i \in \mathcal{U}_t : \hat{s}_{i,t} \text{ is among the ten largest scores in } \mathcal{U}_t\}. \quad (18)$$

The equal-weight portfolio assigns

$$w_{i,t} = \begin{cases} \frac{1}{10}, & i \in \mathcal{P}_t^{Top10}, \\ 0, & \text{otherwise.} \end{cases} \quad (19)$$

The realized portfolio return over the holding period is then

$$R_{p,t}^{(H)} = \sum_{i \in \mathcal{P}_t^{Top10}} w_{i,t} R_{i,t}^{(H)}. \quad (20)$$

The baseline evaluation uses: Top10 equal-weight portfolio; SPY benchmark; predicted Bottom20 portfolio; random Top10 benchmark based on 1000 repeated random draws. The Bottom20 portfolio is used as the primary bottom-ranked diagnostic because it provides a more stable estimate of poorly ranked stocks than Bottom10.

The Bottom20 diagnostic portfolio is defined analogously as the set of the twenty stocks with the lowest predicted scores:

$$\mathcal{P}_t^{Bottom20} = \{i \in \mathcal{U}_t : \hat{s}_{i,t} \text{ is among the twenty lowest scores in } \mathcal{U}_t\}. \quad (21)$$

Its equal-weight return is

$$R_{b,t}^{(H)} = \frac{1}{20} \sum_{i \in \mathcal{P}_t^{Bottom20}} R_{i,t}^{(H)}. \quad (22)$$

The Top-minus-Bottom spread is

$$R_{TM B,t}^{(H)} = R_{p,t}^{(H)} - R_{b,t}^{(H)}. \quad (23)$$

3.6.2 Random Benchmark

For each rebalance date, random Top10 portfolios are sampled from the available stock universe. The random benchmark return is the average return across random draws. The Top10 model portfolio is then compared with this random benchmark using a one-sample t-test on monthly return differences.

For each rebalance date, $M = 1000$ random Top10 portfolios are sampled. Let $\mathcal{R}_t^{(m)}$ denote the m -th random set of ten stocks. The random benchmark return is

$$R_{rand,t}^{(H)} = \frac{1}{M} \sum_{m=1}^M \left(\frac{1}{10} \sum_{i \in \mathcal{R}_t^{(m)}} R_{i,t}^{(H)} \right). \quad (24)$$

3.6.3 Portfolio Optimization Experiments

The paper tests whether optimized weighting improves the ML-selected Top10 portfolio. The tested strategies are: Equal-weight Top10; Static minimum variance Top10; Static score-aware mean-variance Top10; HMM covariance minimum variance Top10; HMM score-aware mean-variance Top10.

Let $\mathcal{P}_t^{Top10}$ denote the ML-selected Top10 universe at date t , and let Σ_t denote the estimated covariance matrix of returns for these stocks. The static minimum-variance portfolio solves

$$\min_{w_t} w_t^\top \Sigma_t w_t \quad (25)$$

subject to

$$\sum_{i \in \mathcal{P}_t^{Top10}} w_{i,t} = 1, \quad 0 \leq w_{i,t} \leq \bar{w}. \quad (26)$$

For the score-aware mean-variance portfolio, predicted ML scores are transformed into standardized relative expected-return signals $\mu_{i,t}^{score}$. The optimizer solves

$$\max_{w_t} w_t^\top \mu_t^{score} - \lambda w_t^\top \Sigma_t w_t \quad (27)$$

subject to

$$\sum_{i \in \mathcal{P}_t^{Top10}} w_{i,t} = 1, \quad 0 \leq w_{i,t} \leq \bar{w}. \quad (28)$$

The parameter $\lambda > 0$ controls the trade-off between exposure to the model score and portfolio risk.

In the empirical implementation, $\bar{w} = 0.25$.

The optimizers are applied only after the ML model has selected the Top10 stocks. This isolates the portfolio construction question from the forecasting question. The score-aware mean-variance objective uses standardized ML scores as relative expected-return signals rather than treating raw model outputs as calibrated expected returns. Covariance matrices are estimated using trailing daily returns, with shrinkage and maximum weight constraints for stability.

To reduce estimation error, the sample covariance matrix is shrunk toward its diagonal:

$$\hat{\Sigma}_t^{shrink} = (1 - \delta) \hat{\Sigma}_t + \delta \text{diag}(\hat{\Sigma}_t), \quad (29)$$

where $\delta \in [0, 1]$ is the shrinkage intensity.

For the HMM regime-aware covariance portfolios, the covariance matrix is estimated using only past observations assigned to the current predicted regime:

$$\hat{\Sigma}_t^{HMM} = \text{Cov} \left(r_\tau : \tau < t, \hat{Z}_\tau = \hat{Z}_t \right). \quad (30)$$

If the number of same-regime observations is below the minimum threshold, the estimator falls back to the standard trailing covariance matrix.

3.7 Experimental Design

The empirical design is built around a frozen baseline. Once the final baseline is selected, all further tests are treated as robustness checks or controlled variations. The final baseline is: Bloomberg-adjusted U.S. equity data; technical, momentum, fundamental, and cross-asset features; 20-trading-day future rank-percentile target; XGBRegressor ensemble over seven seeds; monthly equal-weight Top10 portfolio; Bottom20 diagnostic; SPY and random Top10 benchmarks; 2021–2025 out-of-sample test. The experiments are: Feature ablation and feature analyses; Hyperparameter optimization; Portfolio concentration tests; Horizon and rebalancing tests; Point-In-Time tests; Mega-cap removal; HMM regime forecasting tests; Regime-conditional performance analysis; Portfolio optimization tests; Transaction cost robustness. Each experiment changes only one methodological component at a time where possible.

As an additional robustness check, the XGBoost ranking model is also evaluated under an expanding-window walk-forward retraining scheme. At each monthly rebalance date t , the model is retrained using all observations whose 20-trading-day future returns would have been known by that date. In practice, this means that the training sample includes observations dated no later than $t - H$, where $H = 20$ trading days is the prediction horizon. This avoids look-ahead bias while allowing the ranking model to incorporate newly realized information during the 2021–2025 test period. The hyperparameters and feature set are kept fixed, so the experiment tests retraining rather than additional model selection. Results are reported in section 4.9

3.8 Statistical Evaluation Framework

The model is evaluated using both predictive and portfolio metrics.

3.8.1 Predictive Metrics

The main predictive metric is the Spearman information coefficient, computed cross-sectionally at each rebalance date between predicted scores and realized future returns. The average IC is tested using a one-sample t-test. Pearson correlation and R^2 are also reported, but they are less central because the task is ranking rather than calibrated return prediction.

At each rebalance date, the cross-sectional Spearman information coefficient is

$$IC_t = \rho_S \left(\hat{s}_{i,t}, R_{i,t}^{(H)} \right)_{i \in \mathcal{U}_t}, \quad (31)$$

where $\rho_S(\cdot, \cdot)$ denotes the Spearman rank correlation. The average information coefficient is

$$\overline{IC} = \frac{1}{T} \sum_{t=1}^T IC_t. \quad (32)$$

For decile analysis, stocks are sorted into ten groups according to $\hat{s}_{i,t}$. Let $\mathcal{D}_{q,t}$ denote the set of stocks in predicted-score decile q at time t . The equal-weight decile return is

$$R_{q,t}^{(H)} = \frac{1}{|\mathcal{D}_{q,t}|} \sum_{i \in \mathcal{D}_{q,t}} R_{i,t}^{(H)}. \quad (33)$$

3.8.2 Portfolio Metrics

Portfolio performance is evaluated using: total return; CAGR; annualized arithmetic return; annualized volatility; Sharpe ratio; maximum drawdown; beta versus SPY; annualized alpha versus SPY; annualized excess return versus SPY. The Sharpe ratio uses a risk-free rate of 4%. This value is used as a simple approximation to the short-term U.S. risk-free rate during the

2021–2025 test period. The assumption affects only the Sharpe ratio and does not affect portfolio returns, rankings, or the return-spread significance tests.

Let $R_{p,t}$ denote the monthly portfolio return and let T denote the number of monthly test observations. Cumulative wealth is

$$W_T = \prod_{t=1}^T (1 + R_{p,t}). \quad (34)$$

The total return is

$$TR = W_T - 1. \quad (35)$$

The compound annual growth rate is

$$CAGR = W_T^{\frac{12}{T}} - 1. \quad (36)$$

Annualized volatility is computed as

$$\sigma_{ann} = \sqrt{12} \text{sd}(R_{p,t}). \quad (37)$$

The annualized arithmetic return is

$$\bar{R}_{ann} = 12 \times \bar{R}_p. \quad (38)$$

The Sharpe ratio is

$$SR = \frac{\bar{R}_{ann} - R_f}{\sigma_{ann}}, \quad (39)$$

where R_f is the annual risk-free rate (4%).

Alpha is computed as the annualized intercept implied by the strategy's average return after adjusting for beta exposure to SPY: The strategy beta relative to SPY is estimated as

$$\beta_p = \frac{\text{Cov}(R_{p,t}, R_{m,t})}{\text{Var}(R_{m,t})}, \quad (40)$$

where $R_{m,t}$ is the SPY return.

Annualized alpha is then computed as

$$\alpha_p = 12 (\bar{R}_p - \beta_p \bar{R}_m). \quad (41)$$

annualized according to the monthly evaluation frequency.

Maximum drawdown is computed from the cumulative wealth process as

$$MDD = \min_{1 \leq t \leq T} \left(\frac{W_t}{\max_{1 \leq s \leq t} W_s} - 1 \right). \quad (42)$$

As a robustness check, alpha is also evaluated using standard multi-factor asset-pricing models. This is important because the SHAP analysis indicates that firm size, valuation, profitability, and momentum-related variables are important predictors in the ranking model. The main multi-factor specification is

$$R_{p,t} - R_{f,t} = \alpha + \beta_M(MKT_t - RF_t) + \beta_S SMB_t + \beta_H HML_t + \beta_R RMW_t + \beta_C CMA_t + \beta_U MOM_t + \varepsilon_t. \quad (43)$$

The factors are obtained from Kenneth French's Data Library. The five-factor model controls for market, size, value, profitability, and investment factors, and the momentum factor is added separately. Because the portfolio is evaluated over 20-trading-day holding periods, daily factor returns are compounded over the same holding windows as the strategy returns before estimating the regressions. Standard errors are computed using HAC/Newey-West corrections.

3.8.3 Statistical Tests

The following p-values are reported: Top10 minus Bottom20; Top10 minus RandomTop10; Top10 minus SPY; Spearman IC. All are computed using one-sample t-tests over monthly observations.

For each return spread series d_t , such as Top10-minus-SPY or Top10-minus-Bottom20, the null hypothesis is

$$H_0 : \mathbb{E}[d_t] = 0. \quad (44)$$

The one-sample t-statistic is

$$t = \frac{\bar{d}}{s_d / \sqrt{T}}, \quad (45)$$

where $\bar{d}$ is the sample mean of the spread, s_d is its sample standard deviation, and T is the number of monthly observations.

Because the empirical analysis reports several related statistical tests across alternative models, feature sets, horizons, portfolio concentration levels, regime-aware specifications, portfolio optimization methods, transaction-cost assumptions, and factor-adjusted alpha regressions, the paper also applies multiple-testing corrections. The main correction is the Benjamini-Hochberg false discovery rate procedure, applied within families of related tests. This choice is less conservative than Bonferroni and is appropriate because many robustness tests are correlated variants of the same underlying strategy. Holm and Bonferroni corrections are also reported as conservative family-wise error-rate checks. The main pre-specified baseline diagnostics are treated as one family, while exploratory robustness tests are grouped by experiment type.

In addition to testing whether individual return spreads are different from zero, the paper also tests whether alternative specifications perform differently from the final baseline. Because all specifications are evaluated on the same monthly rebalance dates, these comparisons are conducted using paired tests. For an alternative specification A and the baseline specification B , the paired difference is defined as

$$d_t^{A-B} = S_t^A - S_t^B, \quad (46)$$

where S_t denotes the relevant monthly performance series, such as the Top10 return, the Top10-minus-SPY spread, the Top-minus-Bottom spread, or the Top-minus-Random spread. The null hypothesis is

$$H_0 : \mathbb{E} \left[d_t^{A-B} \right] = 0. \quad (47)$$

The main reported test is a paired mean test estimated as a regression of d_t^{A-B} on a constant, with HAC/Newey-West standard errors. This approach tests whether an alternative specification is statistically different from the baseline rather than relying only on side-by-side comparisons of CAGR, Sharpe ratios, or individual p-values.

The paired-test framework is used whenever an alternative specification is compared directly with the final baseline. This includes the regime-aware models, alternative machine-learning models, portfolio-optimization methods, and the walk-forward retraining specification. These tests answer a different question from the individual benchmark p-values: they test whether an alternative specification improves or deteriorates relative to the baseline on the same monthly rebalance dates.

4 Empirical Results

4.1 Baseline Ranking Performance

The final baseline achieves strong out-of-sample performance over 60 monthly test periods from 2021 to 2025.

Table 1: Out-of-sample portfolio performance, 2021–2025

Portfolio	Total return	CAGR	Annual return	Volatility	Sharpe	Max drawdown	Excess vs SPY	Beta vs SPY	SPY alpha	FF5+MOM alpha
RankModel Top10	3.3594	34.24%	32.10%	21.97%	1.2787	-12.27%	17.24%	1.2489	13.55%	17.68%
SPY	0.9866	14.72%	14.85%	14.54%	0.7463	-20.47%	0.00%	1.0169	-0.25%	–
Bottom20 Predicted	0.8885	13.56%	13.97%	15.68%	0.6363	-19.14%	-0.88%	0.7788	2.41%	–
TopMinusBottom	1.2531	17.64%	18.12%	19.36%	0.7293	-15.33%	–	–	–	–

Notes: Performance is computed over 60 monthly rebalance periods from 2021 to 2025. RankModel Top10 is the implementable equal-weight portfolio of the ten highest-ranked stocks. SPY alpha is the single-factor alpha computed relative to SPY. FF5+MOM alpha is the annualized intercept from a multi-factor regression of the Top10 portfolio excess return on market, size, value, profitability, investment, and momentum factors. The FF5+MOM alpha is statistically significant, with $t(\alpha) = 2.576$ and $p(\alpha) = 0.0100$, using HAC/Newey-West standard errors.

The final column addresses the concern that the strategy’s SPY-adjusted alpha may reflect exposure to standard equity factors. After controlling for market, size, value, profitability, investment, and momentum factors, the Top10 portfolio still has a positive annualized alpha of 17.68%, with a p -value of 0.0100. This suggests that the baseline result is not explained away by conventional factor exposures.

The main ranking diagnostics are:

Table 2: Ranking diagnostics for the final baseline model

Diagnostic	Value
Average Top10 monthly return	2.67%
Average Bottom20 monthly return	1.16%
Average Top-Bottom spread	1.51%
Annualized Top-Bottom spread	18.12%
Top-Bottom p -value	0.0407
Annualized Top10 excess vs Random	16.30%
Top-Random p -value	0.0112
Annualized Top10 excess vs SPY	17.24%
Top-SPY p -value	0.0050
Average Spearman IC	0.0424
Spearman IC p -value	0.0464
Percent months Top beats SPY	66.67%
Percent positive IC	60.00%

Notes: The ranking diagnostics evaluate whether high predicted scores correspond to higher realized future returns. P-values are computed from one-sample tests on monthly return spreads or monthly information coefficients.

The strategy therefore produces both economically meaningful returns and statistically significant ranking performance.

To complement the numerical performance statistics, Figures 3–5 provide visual evidence on cumulative performance, cross-sectional ranking quality, and year-by-year robustness.

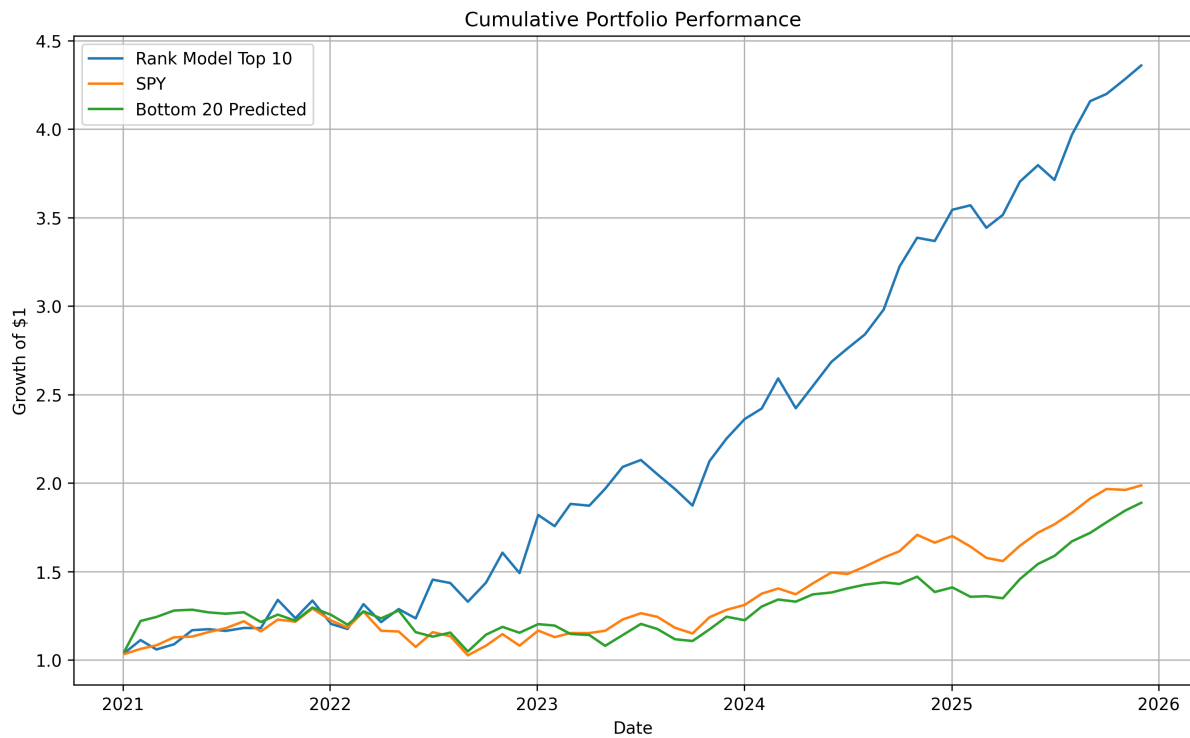


Figure 3: Cumulative out-of-sample portfolio performance

Figure 3 reports the cumulative growth of one dollar invested in the RankModel Top10 portfolio, SPY, and the predicted

Bottom20 portfolio during the 2021–2025 test period. The figure illustrates that the Top10 portfolio begins to separate materially from both benchmarks from 2023 onward and compounds to substantially higher terminal wealth by the end of the sample. The Bottom20 portfolio remains much closer to SPY, which supports the interpretation that the model’s strongest economic value comes from identifying high-ranked winners rather than simply avoiding the lowest-ranked stocks.

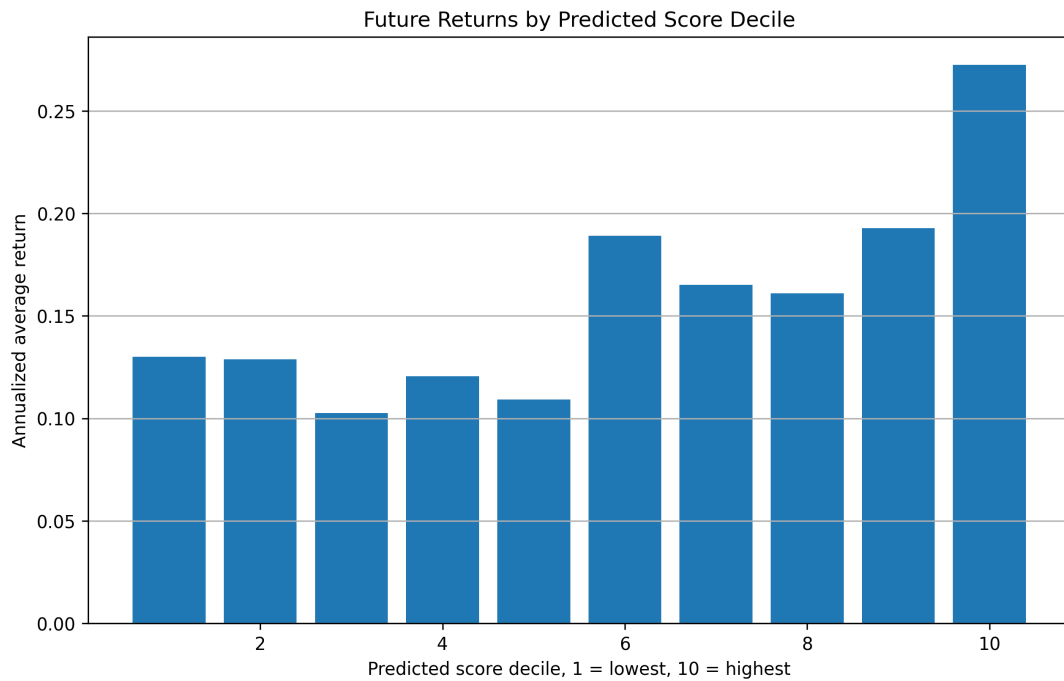


Figure 4: Average future returns by predicted score decile

Figure 4 shows the annualized average future return of stocks sorted into deciles according to their predicted ranking score, where decile 1 contains one of the lowest-ranked stocks and decile 10 contains the highest-ranked stocks. The figure provides a cross-sectional validation of the ranking model beyond the Top10 portfolio alone. Although the relationship is not perfectly monotonic, the highest predicted decile delivers the strongest average future return, suggesting that the model assigns meaningfully higher scores to stocks with superior subsequent performance.

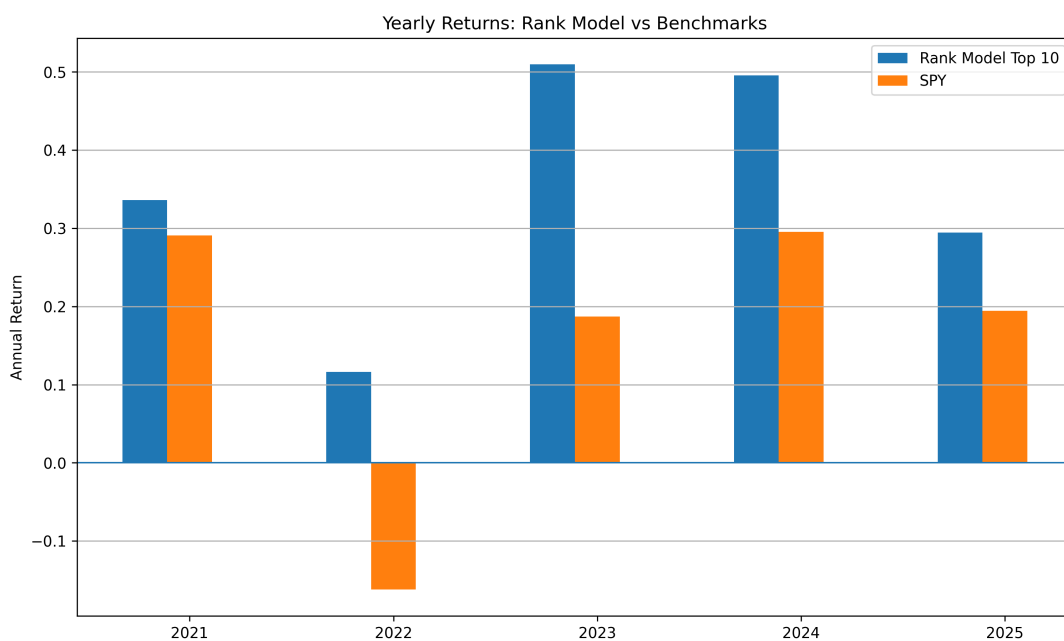


Figure 5: Yearly returns of the RankModel Top10 portfolio and SPY

Figure 5 reports calendar-year returns for the RankModel Top10 portfolio and SPY over the out-of-sample period. The Top10 portfolio outperforms SPY in each test year shown, including 2022, when SPY generated a negative return while the strategy remained positive. This yearly decomposition indicates that the aggregate outperformance is not driven by a single isolated month or year, but reflects persistent performance across different market environments.

4.2 Robustness to Model Choice

Table 3: Alternative machine learning model comparison

Model	CAGR	Sharpe	Top-Bottom p	Top-Random p	Top-SPY p	Spearman IC p
XGBoost	34.24%	1.279	0.0407	0.0112	0.0050	0.0464
Random Forest	33.37%	1.222	0.0391	0.0291	0.0092	0.0943
LightGBM	31.95%	1.249	0.0562	0.0141	0.0062	0.1025
Extra Trees	28.08%	1.030	0.0541	0.1068	0.0684	0.0782
Ridge Regression	15.54%	0.678	0.8896	0.9671	0.8368	0.2560
Neural Network	16.23%	0.679	0.8948	0.8000	0.7010	0.6201

Notes: All models are evaluated using the same feature set, target variable, train-validation-test split, monthly Top10 portfolio construction, and statistical evaluation procedure. XGBoost corresponds to the final baseline model. The alternative models are not extensively hyperparameter-optimized; they are included to compare broad model classes under a common empirical protocol.

Table 3 shows that the main empirical result is not unique to one tree-based algorithm. Random Forest and LightGBM both produce economically strong Top10 portfolios, with CAGRs of 33.37% and 31.95%, respectively, and statistically significant outperformance relative to SPY. This suggests that nonlinear tree-based models are well suited to the stock-ranking problem. However, XGBoost remains the strongest overall specification because it delivers the highest CAGR, the strongest Sharpe ratio among the main tree-based models, and the most balanced statistical evidence across the Top-Bottom, Top-Random, Top-SPY, and Spearman IC diagnostics.

The contrast with Ridge regression is particularly informative. Ridge produces a CAGR of only 15.54%, a Sharpe ratio of 0.678, and no statistically significant ranking or portfolio evidence. This indicates that a linear specification is unable to capture the nonlinear interactions among momentum, relative strength, volatility, fundamentals, and cross-asset variables that appear to drive the ranking signal. The model comparison therefore supports the use of nonlinear ensemble methods, while justifying the selection of XGBoost as the final baseline.

To complement the individual model diagnostics, Table 4 compares each alternative model directly with the final XGBoost baseline using paired tests on the same monthly rebalance dates. This evaluates whether an alternative model significantly improves or deteriorates relative to the selected baseline, rather than only whether it performs well in absolute terms.

Table 4: Paired tests: alternative machine-learning models against XGBoost baseline

Alternative model	Top10 return		Top-SPY		Top-Bottom		Top-Random	
	Ann. diff.	HAC p	Ann. diff.	HAC p	Ann. diff.	HAC p	Ann. diff.	HAC p
Random Forest	-0.56%	0.9002	-0.56%	0.9002	1.88%	0.7436	-0.60%	0.8931
LightGBM	-1.97%	0.4209	-1.97%	0.4209	-2.22%	0.5082	-1.91%	0.4360
Extra Trees	-4.64%	0.4468	-4.64%	0.4468	-0.40%	0.9654	-4.59%	0.4533
Ridge Regression	-16.05%	0.0210	-16.05%	0.0210	-17.18%	0.0801	-16.09%	0.0208
Neural Network	-15.20%	0.0140	-15.20%	0.0140	-18.99%	0.0447	-15.07%	0.0149

Notes: The table reports paired tests comparing each alternative model with the final XGBoost baseline on the same 60 monthly rebalance dates. The annualized difference equals twelve times the mean monthly paired difference between the alternative model and the baseline. HAC p -values use Newey-West standard errors with three lags. A negative difference indicates underperformance relative to XGBoost.

Table 4 shows that Random Forest, LightGBM, and Extra Trees are not statistically different from the XGBoost baseline in paired tests. This supports the interpretation that the result is not unique to one exact tree-based algorithm. By contrast, Ridge regression and the neural network are significantly weaker than XGBoost for Top10 returns, Top-SPY performance,

and Top-Random performance. The paired tests therefore reinforce the conclusion that nonlinear tree-based methods are important, while XGBoost remains the preferred baseline specification.

4.2.1 Hyperparameter Search and Early-Stopping Robustness

Randomized hyperparameter searches were conducted using validation and walk-forward validation procedures. None of the optimized configurations improved the final out-of-sample result. The best reduced hyperparameter search result achieved 31.63% CAGR, but weakened key statistical diagnostics, including Top-Bottom $p = 0.1068$, Top-Random $p = 0.0564$, Top-SPY $p = 0.0230$, and Spearman IC $p = 0.0981$. The original XGBoost specification is therefore retained because it provides the best balance of out-of-sample performance, statistical significance, and methodological stability.

Table 5: Hyperparameter search robustness

Specification	CAGR	Top-Bottom p	Top-Random p	Top-SPY p	Spearman IC p
Final baseline XGBoost	34.24%	0.0407	0.0112	0.0050	0.0464
Best reduced hyperparameter search	31.63%	0.1068	0.0564	0.0230	0.0981

Notes: The hyperparameter search was conducted using validation and walk-forward validation procedures. The optimized configuration did not improve the final out-of-sample performance or the main ranking diagnostics. Sharpe ratio for the best reduced hyperparameter-search specification was not retained in the final results record.

As an additional overfitting robustness check, the baseline XGBoost model was re-estimated with a large maximum number of trees and validation-based early stopping. Specifically, the model was allowed to fit up to 2,000 trees, with early stopping after 50 validation rounds without improvement. This specification achieved a CAGR of 33.64% and remained statistically significant relative to SPY, with a Top-SPY p -value of 0.0080, and relative to the random benchmark, with a Top-Random p -value of 0.0301. However, it weakened the Sharpe ratio, the Top-Bottom test, and the Spearman IC test relative to the original baseline. The fixed 600-tree specification is therefore retained as the final baseline because it provides stronger overall statistical evidence and a better balance between economic performance and ranking quality.

Table 6: Early-stopping robustness check

Specification	CAGR	Sharpe	Top-Bottom p	Top-Random p	Top-SPY p	Spearman IC p
Final baseline XGBoost	34.24%	1.279	0.0407	0.0112	0.0050	0.0464
Early-stopped XGBoost	33.64%	1.204	0.0547	0.0301	0.0080	0.0859

Notes: The early-stopped specification uses a maximum of 2,000 trees and stops training after 50 validation rounds without improvement. The comparison is based on the same train-validation-test split and Top10 portfolio construction. The Top-Bottom diagnostic in the early-stopped run uses Bottom10 rather than Bottom20, so it should be interpreted as a robustness check rather than an exact replacement for the main Top10/Bottom20 baseline.

4.3 Multi-Factor Alpha Robustness

A natural concern is that the strategy's SPY-adjusted alpha may reflect exposure to known equity factors rather than genuine abnormal performance. This concern is especially relevant because the SHAP analysis shows that firm size, valuation, profitability, and momentum-related predictors are important in the XGBoost model. To address this issue, the Top10 portfolio is regressed on standard asset-pricing factors.

Table 7: Multi-factor alpha robustness

Model	Ann. alpha	$t(\alpha)$	$p(\alpha)$	Adj. R^2	MKT	SMB	HML	RMW	CMA	MOM
CAPM	16.34%	2.859	0.0043	0.677	1.216	–	–	–	–	–
FF3	18.01%	3.040	0.0024	0.674	1.179	0.187	-0.024	–	–	–
Carhart4	18.54%	2.894	0.0038	0.671	1.154	0.167	-0.037	–	–	-0.089
FF5	16.93%	2.919	0.0035	0.671	1.168	0.112	0.126	-0.046	-0.280	–
FF5 + MOM	17.68%	2.576	0.0100	0.667	1.146	0.081	0.105	-0.093	-0.245	-0.079

Notes: The dependent variable is the Top10 portfolio excess return. Daily Fama-French factor returns are compounded over the same 20-trading-day holding windows as the portfolio returns. CAPM includes the market factor only. FF3 includes market, size, and value. Carhart4 adds momentum. FF5 includes market, size, value, profitability, and investment. FF5 + MOM adds momentum to the five-factor model. Alpha is annualized by multiplying the holding-period intercept by 12. Standard errors use HAC/Newey-West corrections. The regressions use 59 holding-period observations rather than 60 because daily factor returns are compounded over complete 20-trading-day holding windows, causing one monthly observation to be lost during factor-return alignment.

Table 8: Factor-loading significance in the FF5 + momentum regression

Factor	Coefficient	t -statistic	p -value
MKT	1.146	13.437	< 0.0001
SMB	0.081	0.423	0.6725
HML	0.105	0.614	0.5394
RMW	-0.093	-0.420	0.6748
CMA	-0.245	-1.239	0.2153
MOM	-0.079	-0.363	0.7166

Notes: The table reports coefficient estimates, HAC/Newey-West t -statistics, and p -values for the full FF5 + momentum specification. The dependent variable is the Top10 portfolio excess return. The regression uses 59 holding-period observations rather than 60 because daily factor returns are compounded over complete 20-trading-day holding windows. The final rebalance date, 2025-12-01, does not have a complete subsequent 20-trading-day factor window and is therefore excluded from the factor-regression sample.

Table 7 shows that the Top10 portfolio’s abnormal return is not explained away by standard equity factors. The annualized alpha remains positive and statistically significant across all specifications, including the FF5 + momentum model, where the annualized alpha is 17.68% with a p -value of 0.0100. The portfolio has a market beta above one. Table 8 shows that the market loading is highly significant, but the size, value, profitability, investment, and momentum loadings are not individually significant in the full FF5 + momentum specification. This suggests that the strategy is not merely a disguised exposure to conventional size, value, profitability, investment, or momentum premia; instead, the machine-learning ranking model appears to add incremental stock-selection value beyond these standard factors.

4.4 Feature Importance and Ablation

Feature ablation shows that cross-asset features improve performance, while rates and slow-moving macro-cycle variables degrade performance.

Cross-asset market variables appear to provide useful contemporaneous information about market conditions. In contrast, macro-cycle variables such as CPI, unemployment, PMI, consumer confidence, and Fed Funds add noise and reduce ranking quality.

Table 9: Feature ablation results

Feature set	CAGR	Sharpe	Top-Bottom p	Top-Random p	Top-SPY p	Spearman IC p
Original baseline	31.27%	1.227	0.0434	0.0094	0.0040	0.0492
Original + cross-asset	34.24%	1.279	0.0407	0.0112	0.0050	0.0464
Original + cross-asset + rates	32.17%	1.223	0.1081	0.0172	0.0202	0.1231
Original + cross-asset + macro-cycle	22.38%	0.935	0.5997	0.1852	0.1546	0.1490

Notes: The final baseline corresponds to the “Original + cross-asset” specification. Cross-asset variables include bond, dollar, gold, and oil returns. Rates and slow-moving macro-cycle variables were tested but excluded from the final model because they weakened out-of-sample ranking and portfolio performance.

4.4.1 Feature Importance and Model Interpretation (SHAP)

To improve the interpretability of the XGBoost model, SHAP (SHapley Additive exPlanations) values are computed on the test set. Unlike the built-in feature importance of tree-based models, SHAP values quantify the average contribution of each predictor to the model’s output while accounting for interactions among features.

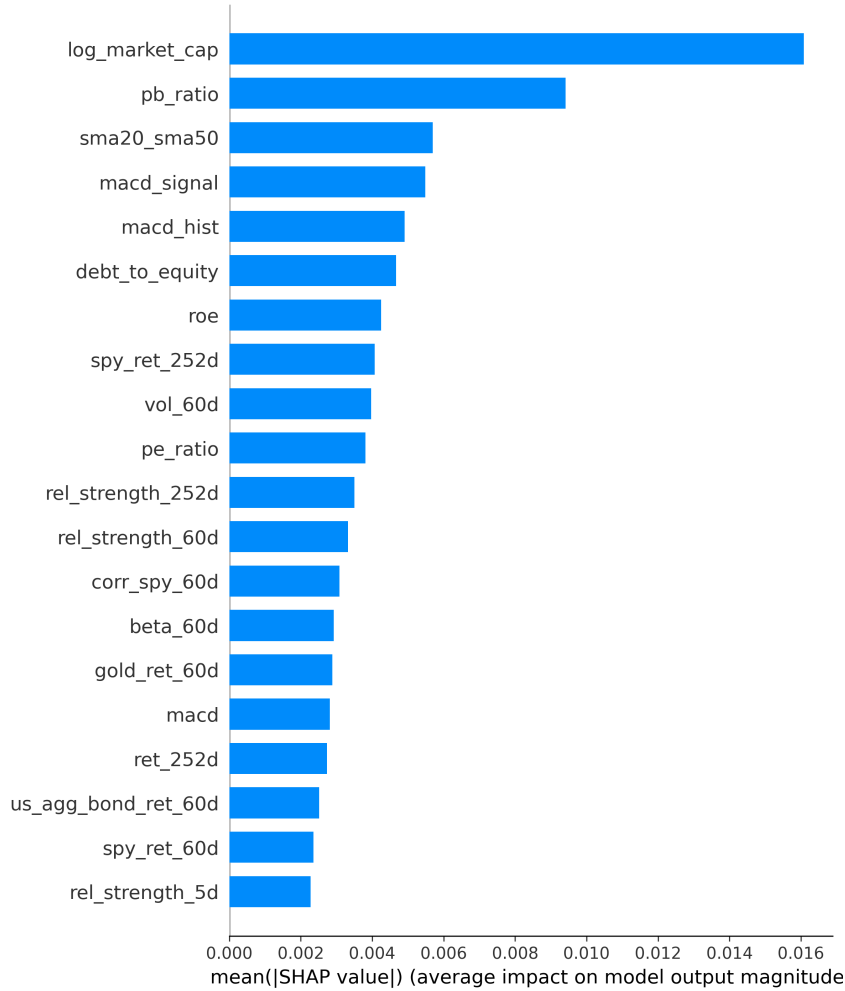


Figure 6: SHAP Importance

Figure 6 presents the global feature importance measured by the mean absolute SHAP value. Firm size (log market capitalization) is the dominant predictor, followed by valuation ratios (price-to-book and price-to-earnings), technical indicators (MACD and moving-average ratios), profitability (return on equity), leverage (debt-to-equity), and long-term momentum variables. Cross-asset features such as gold and aggregate bond returns also contribute to the model, indicating that both firm-specific and macro-financial information influence stock selection.

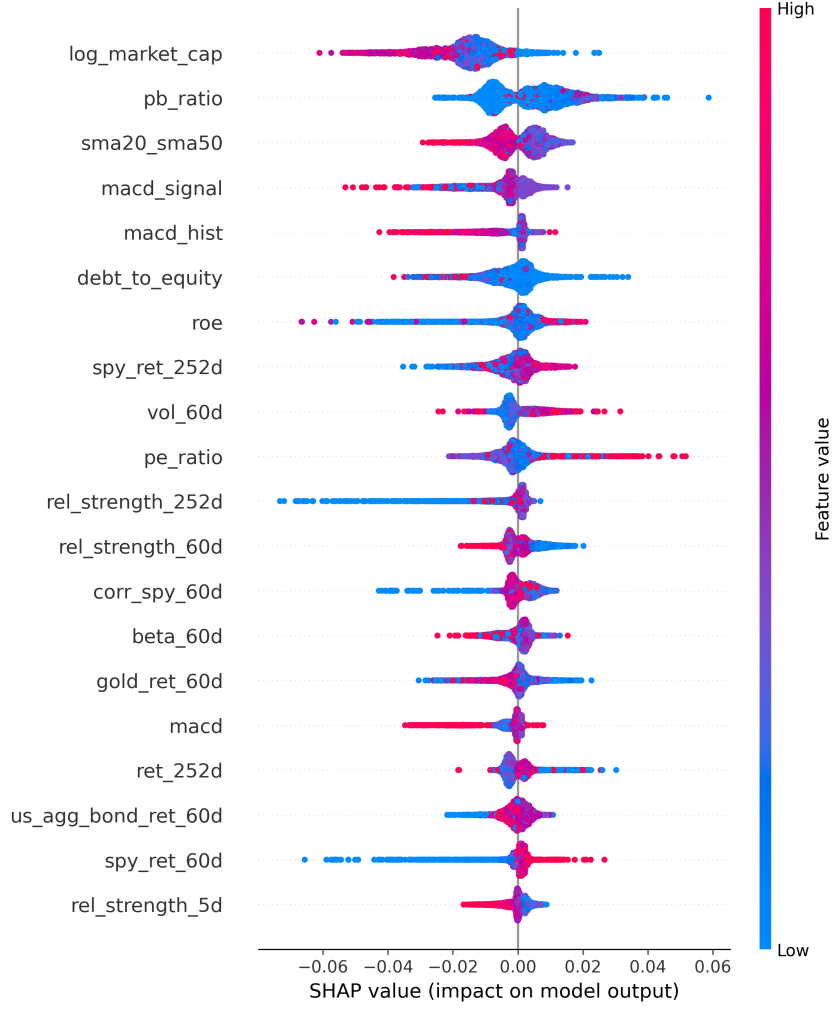


Figure 7: SHAP Summary

Figure 7 further illustrates the direction of these effects. For example, smaller firms generally contribute positively to the predicted return score, consistent with the size effect documented in the empirical asset pricing literature. Overall, the SHAP analysis suggests that the model combines valuation, quality, momentum, technical, and macroeconomic information rather than relying on a single class of predictors.

The SHAP analysis corroborates the feature ablation results. Four of the five accounting-based variables appear among the ten most influential predictors, indicating that the improvement in portfolio performance observed after incorporating fundamentals is driven by economically meaningful information rather than by overfitting or spurious correlations.

4.5 Regime-Aware Forecasting

4.5.1 HMM Regime as an Additional Feature

The first regime experiment adds the current HMM regime as a one-hot encoded feature to the global XGBoost model. This tests whether the model can use regime information interactively with technical, fundamental, and cross-asset variables.

When regimes are used as features, the predicted regime $\hat{Z}_t$ is converted into dummy variables. For each regime $k = 1, \dots, K$, define

$$D_{k,t} = \begin{cases} 1, & \text{if } \hat{Z}_t = k, \\ 0, & \text{otherwise.} \end{cases} \quad (48)$$

The augmented feature vector is then

$$\tilde{X}_{i,t} = [X_{i,t}, D_{1,t}, D_{2,t}, \dots, D_{K,t}]. \quad (49)$$

This approach produces materially lower economic performance (paired tests in Section 4.5.4 discuss this further):

Table 10: Effect of adding the HMM regime as a predictive feature

Model	CAGR	Sharpe	Top-Bottom p	Top-Random p	Top-SPY p	Spearman IC p
Baseline	34.24%	1.279	0.0407	0.0112	0.0050	0.0464
HMM as feature	30.37%	1.120	0.0868	0.0426	0.0210	0.0433

Notes: The HMM feature specification adds one-hot encoded expanding-window HMM regime indicators to the baseline XGBoost feature set. Portfolio performance is evaluated on the monthly Top10 strategy over the 2021–2025 test period.

The HMM feature slightly improves the Spearman IC p-value but reduces portfolio performance and weakens the Top-Bottom p-value. This suggests that regime information contains some predictive content but does not improve the economic ranking sufficiently to enhance the Top10 portfolio.

4.5.2 One Model per Regime

The second regime experiment trains one XGBoost model per HMM regime. At prediction time, the current regime determines which model is used.

In the regime-specific specification, a separate forecasting function is estimated for each regime:

$$\hat{f}_k = \arg \min_{f \in \mathcal{F}} \sum_{(i,t): \hat{Z}_t = k} (y_{i,t} - f(X_{i,t}))^2 + \Omega(f), \quad k = 1, \dots, K. \quad (50)$$

The summation $\sum_{(i,t): \hat{Z}_t = k}$ is taken only over observations for which the estimated HMM regime is k , where $\hat{Z}_t$ denotes the inferred market regime at time t . Consequently, each XGBoost model is trained exclusively on observations belonging to its corresponding market regime. The variable $y_{i,t}$ represents the target variable, while $X_{i,t}$ denotes the vector of explanatory variables or predictive features. The squared loss term, $(y_{i,t} - f(X_{i,t}))^2$, measures the prediction error for each observation, and the optimization seeks to minimize the total prediction error across all observations assigned to regime k . The regularization term $\Omega(f)$ is the standard XGBoost complexity penalty, which discourages overly complex tree structures and reduces the risk of overfitting.

At prediction time, the score assigned to stock i is

$$\hat{s}_{i,t} = \hat{f}_{\hat{Z}_t}(X_{i,t}). \quad (51)$$

The result is economically weaker than the baseline (paired tests in Section 4.5.4 discuss this further):

Table 11: Performance of regime-specific XGBoost models

Model	CAGR	Sharpe	Top-Bottom p	Top-Random p	Top-SPY p	Spearman IC p
Baseline	34.24%	1.279	0.0407	0.0112	0.0050	0.0464
One model per HMM regime	23.44%	0.905	0.1086	0.1631	0.1562	0.3472

Notes: The regime-specific specification trains a separate XGBoost model within each predicted HMM regime. At each test date, the model corresponding to the current predicted regime is used to rank stocks.

The likely explanation is sample fragmentation. Each regime-specific model is trained on only a subset of the training observations, reducing effective sample size and increasing overfitting risk. The global model benefits from a much larger training sample and appears to learn more robust relationships.

4.5.3 Regime-Conditional Performance of the Baseline

Although regimes do not improve aggregate forecasting, they are useful for interpretation. The baseline model performs differently across HMM regimes.

Table 12: Baseline strategy performance conditional on expanding-window HMM regime

Regime	Economic interpretation	Test months	CAGR	Sharpe	Average IC
Regime 0	Normal Expansion	19	32.5%	1.40	0.052
Regime 1	Bull Market	16	47.7%	1.80	0.053
Regime 2	Correction / Bear Market	13	39.6%	1.16	0.011
Regime 3	Crisis / Extreme Stress	12	15.6%	0.58	0.048

Notes: The table reports the performance of the baseline equal-weight Top10 strategy conditional on the expanding-window HMM regime prevailing at each monthly rebalance date. Economic interpretations are assigned using the expanding-window HMM regime-characteristic table, based on average return, volatility, momentum, drawdown, VIX, and yield-spread behavior.

Regime 1 is interpreted as a bull or high-return expansion state because its regime-characteristic statistics show the strongest annualized market return and positive momentum without crisis-level volatility or drawdown. Regime 3 is interpreted as a crisis or extreme-stress state because it has the weakest momentum, the deepest worst drawdown, and the highest composite stress score. The strategy performs best in Regime 1 and worst in Regime 3, indicating that the model is most effective in constructive market environments and least effective during severe stress. This supports the conclusion that regimes help explain when the model performs well, even though direct regime-aware forecasting does not improve aggregate results.

4.5.4 Formal Paired Tests Between Specifications

To avoid relying only on side-by-side comparisons of summary statistics, the regime-aware specifications are also compared with the final baseline using paired tests on monthly return and return-spread series. For each alternative specification, the paired difference is computed on the same monthly rebalance dates as the baseline. A negative value indicates that the alternative specification underperforms the baseline on average.

Table 13: Paired tests against the final baseline

Alternative specification	Top10 return		Top-SPY		Top-Bottom		Top-Random	
	Ann. diff.	HAC p	Ann. diff.	HAC p	Ann. diff.	HAC p	Ann. diff.	HAC p
HMM as feature	-2.89%	0.1732	-2.89%	0.1732	-3.11%	0.1548	-2.86%	0.1745
One model per HMM regime	-8.62%	0.0838	-8.62%	0.0838	-6.09%	0.5216	-8.71%	0.0837

Notes: The table reports paired tests comparing each regime-aware specification with the final baseline on the same 60 monthly rebalance dates. The annualized difference equals twelve times the mean monthly paired difference between the alternative and the baseline. Top-SPY, Top-Bottom, and Top-Random refer to paired differences in the corresponding return-spread series. HAC p -values use Newey-West standard errors with three lags. A negative annualized difference indicates that the alternative specification underperforms the baseline on average.

Table 13 shows that both regime-aware forecasting specifications underperform the baseline economically. The HMM-as-feature model has lower Top10, Top-SPY, Top-Bottom, and Top-Random performance, but none of the paired differences is statistically significant at conventional levels. The one-model-per-regime specification shows larger negative differences, with Top10, Top-SPY, and Top-Random underperformance marginally significant at the 10% level but not at the 5% level. These results provide no evidence that regime-aware forecasting improves the baseline ranking model. At the same time, they suggest that the weaker performance of the regime-aware alternatives should be interpreted cautiously: the alternatives are economically less attractive, but not always statistically significantly worse.

4.6 Portfolio Optimization

The portfolio optimization experiments show that equal weighting remains the best portfolio construction method.

The optimized portfolios reduce or reshape risk but do not improve the risk-adjusted performance of the equal-weight strategy. This is consistent with the idea that covariance estimation error and score calibration uncertainty can outweigh the theoretical benefits of optimization.

Table 15 reports paired tests comparing each optimized portfolio directly with the equal-weight Top10 baseline. This

Table 14: Portfolio optimization results

Strategy	CAGR	Sharpe	Top-Bottom p	Top-SPY p	SPY-adjusted alpha
EqualWeight Top10	34.24%	1.279	0.0407	0.0050	13.55%
Static MinVar Top10	24.21%	1.088	0.2088	0.1084	9.41%
Static ScoreMV Top10	32.16%	1.074	0.0905	0.0334	10.81%
HMM MinVar Top10	26.54%	1.173	0.1350	0.0596	11.31%
HMM ScoreMV Top10	32.91%	1.095	0.0836	0.0299	11.62%

Notes: Portfolio optimization is applied within the machine-learning-selected Top10 universe. Static strategies use historical covariance estimates, while HMM strategies use regime-dependent covariance estimates. ScoreMV strategies incorporate model scores into a mean-variance objective. Alpha is annualized relative to SPY.

is the relevant comparison because all optimized portfolios are formed within the same machine-learning-selected Top10 universe.

Table 15: Paired tests: optimized portfolios against equal-weight Top10

Alternative strategy	Portfolio return		Top-SPY	
	Ann. diff.	HAC p	Ann. diff.	HAC p
Static MinVar Top10	-8.69%	0.0182	-8.69%	0.0182
Static ScoreMV Top10	-0.86%	0.8692	-0.86%	0.8692
HMM MinVar Top10	-6.73%	0.1044	-6.73%	0.1044
HMM ScoreMV Top10	-0.24%	0.9613	-0.24%	0.9613

Notes: The table reports paired tests comparing each optimized Top10 portfolio with the equal-weight Top10 baseline on the same 60 monthly rebalance dates. The annualized difference equals twelve times the mean monthly paired difference. HAC p -values use Newey-West standard errors with three lags. Since all strategies use the same selected Top10 universe and SPY benchmark, the paired return difference is the direct test of whether optimization improves realized performance.

The paired tests confirm that portfolio optimization does not significantly improve the equal-weight Top10 baseline. Static minimum variance is significantly weaker than equal weighting, with an annualized paired return difference of -8.69% and a HAC p -value of 0.0182. The score-aware and HMM score-aware portfolios are statistically indistinguishable from equal weighting, but they also do not improve it. These results support the interpretation that, after the ranking model has already selected the Top10 stocks, covariance estimation and score calibration error offset the theoretical benefits of optimization.

4.7 Additional Robustness Checks

4.7.1 Portfolio Concentration

The portfolio concentration test compares Top5, Top10, and Top20 selections.

Table 16: Portfolio concentration robustness

Portfolio	CAGR	Sharpe	Top-Bottom p	Top-Random p	Top-SPY p	Spearman IC p
Top5 / Bottom5	41.98%	1.323	0.0606	0.0087	0.0043	0.0464
Top10 / Bottom20	34.24%	1.279	0.0407	0.0112	0.0050	0.0464
Top20 / Bottom20	31.51%	1.359	0.0450	0.0040	0.0012	0.0464

Notes: The table compares alternative portfolio concentration levels using the same baseline XGBoost ranking model. Top5 is more concentrated, Top20 is more diversified, and Top10 is the retained baseline specification. The Spearman IC is unchanged across rows because the underlying monthly ranking scores are the same; only the portfolio construction rule changes.

Top5 generates the highest CAGR but is more concentrated and less statistically stable against the bottom portfolio. Top20 has the highest Sharpe ratio but lower CAGR. Top10 is retained as the main strategy because it provides a balance between concentration, diversification, economic performance, and statistical reliability.

4.7.2 Horizon and Rebalancing Robustness

The baseline uses a 20-trading-day horizon with monthly rebalancing because the prediction horizon and rebalance frequency are aligned. Alternative horizons produce informative robustness results.

Table 17: Horizon and rebalancing robustness

Experiment	CAGR	Sharpe	Top-Bottom p	Top-Random p	Top-SPY p	Spearman IC p
10-day horizon, monthly	11.67%	0.571	0.9637	0.5550	0.6329	0.9488
20-day horizon, monthly	34.24%	1.279	0.0407	0.0112	0.0050	0.0464
60-day horizon, monthly	82.61%	1.865	0.0837	0.0260	0.0028	0.0533

Notes: The baseline specification uses a 20-trading-day prediction horizon with monthly rebalancing.

The 10-day horizon fails, suggesting that the features do not capture very short-term return dynamics, as is common in the literature. The 60-day horizon performs strongly, indicating that the signal may be even more effective at medium-term horizons. However, a 60-day horizon with monthly evaluation introduces overlapping return observations and should be interpreted as predictive-horizon evidence rather than a clean non-overlapping trading strategy.

4.7.3 Point-in-Time Accounting Data

A common concern in empirical asset pricing studies is that accounting variables may inadvertently incorporate information that was not yet publicly available at the portfolio formation date. To address this issue, all accounting variables (ROE and debt-to-equity) were artificially delayed by 90 calendar days before merging with the daily stock panel, approximating conservative point-in-time availability.

Table 18: Robustness to point-in-time accounting data availability

Metric	Baseline	90-day PIT lag
CAGR	34.2%	34.1%
Annual return	31.7%	31.7%
Annual volatility	20.9%	20.2%
Sharpe ratio	1.28	1.37
Maximum drawdown	-12.3%	-11.4%
Alpha vs. SPY	16.0%	14.2%
Annual excess return vs. SPY	16.9%	16.8%
Top-Bottom p -value	0.0407	0.0175
Top-Random p -value	0.0112	0.0053
Top-SPY p -value	0.0050	0.0018
Spearman IC	0.050	0.043
Spearman IC p -value	0.0464	0.0367
Average Top10 hit rate	13.3%	8.8%

Notes: The baseline model uses Bloomberg fundamental variables as downloaded. The robustness specification delays accounting variables by 90 calendar days before merging them with the daily stock panel, approximating conservative point-in-time availability. Portfolio construction, model specification, training period, and evaluation procedure remain unchanged.

Table 18 shows that introducing a conservative 90-day reporting lag to accounting variables has very little impact on portfolio performance or predictive accuracy. The strategy continues to generate a CAGR above 34%, statistically significant Top-Bottom, Top-Random, and Top-SPY spreads, and a significant positive information coefficient. These results suggest that the model's predictive performance is not driven by look-ahead bias in accounting information.

4.7.4 Mega-Cap Removal

Removing 25 mega-cap and superstar stocks reduces raw return but strengthens several statistical diagnostics.

Table 19: Robustness to removing mega-cap stocks

Metric	Full universe	Ex-largest universe
Number of stocks	206	181
CAGR	34.24%	29.64%
Sharpe	1.279	1.329
Top-Bottom p	0.0407	0.0122
Top-Random p	0.0112	0.0014
Top-SPY p	0.0050	0.0147
Spearman IC p	0.0464	0.0333

Notes: The ex-largest universe removes 25 mega-cap and superstar stocks from the investment universe. The test evaluates whether the baseline result is driven only by the largest technology and growth winners.

This supports the conclusion that the model is not merely selecting mega-cap winners. The signal remains economically and statistically meaningful after removing the largest names.

4.7.5 Transaction Costs

Transaction costs are applied using portfolio turnover. Let $w_{i,t}^{new}$ denote the target portfolio weight after rebalancing at date t , and let $w_{i,t}^{old}$ denote the drifted portfolio weight immediately before rebalancing. Turnover is defined as

$$TO_t = \sum_i |w_{i,t}^{new} - w_{i,t}^{old}|. \quad (52)$$

If c denotes the proportional transaction cost rate, the net portfolio return is

$$R_{p,t}^{net} = R_{p,t}^{gross} - c \times TO_t. \quad (53)$$

This formulation does not require assuming a specific initial capital. Because both returns and transaction costs are expressed in proportional terms, the same net return applies to any initial portfolio value. Average monthly turnover is approximately 148.1%.

Table 20: Transaction cost robustness

Cost per trade	CAGR	Sharpe	SPY-adjusted alpha	Top-SPY p	Top-Bottom p
0 bps	34.24%	1.279	13.55%	0.0050	0.0407
5 bps	33.08%	1.238	12.64%	0.0076	0.0513
10 bps	31.92%	1.197	11.74%	0.0113	0.0642
25 bps	28.51%	1.074	9.03%	0.0347	0.1201
50 bps	23.00%	0.870	4.52%	0.1638	0.2921

Notes: Transaction costs are applied using portfolio turnover. Net monthly returns are computed as gross returns minus the cost rate multiplied by turnover. Alpha is annualized relative to SPY. Results remain statistically significant relative to SPY up to 25 basis points per trade, but weaken substantially at 50 basis points.

Table 20 shows that the strategy remains economically attractive under moderate transaction-cost assumptions. The CAGR falls gradually from 34.24% with no costs to 28.51% at 25 basis points per trade, and the Top-SPY test remains statistically significant. At 50 basis points, however, the alpha and statistical significance weaken substantially. This suggests that implementation costs matter, but the main result is robust to realistic cost levels for liquid U.S. large-cap equities.

4.8 Walk-Forward Retraining

As an additional robustness check, we test whether the ranking model benefits from being periodically updated during the out-of-sample period. In this walk-forward specification, the XGBoost ensemble is retrained at each monthly rebalance date using only observations whose 20-trading-day future returns would already have been known. This evaluates whether adaptive retraining improves performance relative to the frozen baseline model trained before the 2021–2025 test period. The feature set and hyperparameters are kept fixed, so the experiment isolates the effect of expanding-window retraining rather than additional model selection.

Table 21: Walk-forward retraining robustness

Specification	CAGR	Sharpe	Top-Bot. p	Top-Rand. p	Top-SPY p	IC p
Frozen baseline	34.24%	1.279	0.0407	0.0112	0.0050	0.0464
Walk-forward retrain	23.02%	0.866	0.4052	0.2440	0.1375	0.3307

Notes: The frozen baseline trains the XGBoost ensemble on 2005–2018 data and evaluates it over 2021–2025. The walk-forward specification retrains the XGBoost ensemble at each monthly rebalance date using only observations whose 20-trading-day future returns would have been known at that date. The feature set and hyperparameters are kept fixed.

Table 21 shows that periodic retraining does not improve the baseline ranking model. The walk-forward specification remains economically positive, with a 23.02% CAGR, but it is weaker than the frozen baseline and loses statistical significance across the main ranking diagnostics. The Top-Bottom, Top-Random, Top-SPY, and Spearman IC tests are no longer significant at conventional levels.

Table 22: Paired test: walk-forward retraining against frozen baseline

Alternative specification	Top10 return		Top-SPY		Top-Bottom		Top-Random	
	Ann. diff.	HAC p	Ann. diff.	HAC p	Ann. diff.	HAC p	Ann. diff.	HAC p
Walk-forward retrain	-8.83%	0.0154	-8.83%	0.0154	-10.63%	0.0211	-8.76%	0.0159

Notes: The table reports paired tests comparing the walk-forward retrained XGBoost model with the frozen 2005–2018 baseline on the same 60 monthly rebalance dates. A negative annualized difference indicates that the walk-forward model underperforms the frozen baseline on average. HAC p -values use Newey-West standard errors with three lags.

Table 22 shows that the weaker walk-forward result is also statistically meaningful in direct paired comparisons. The annualized paired difference is -8.83% for Top10 returns and Top-SPY performance, -10.63% for Top-Bottom performance, and -8.76% for Top-Random performance. The HAC p -values are below 0.05 in all four comparisons. Thus, the walk-forward specification does not simply fail to improve the frozen baseline; in this sample, it performs significantly worse in paired tests.

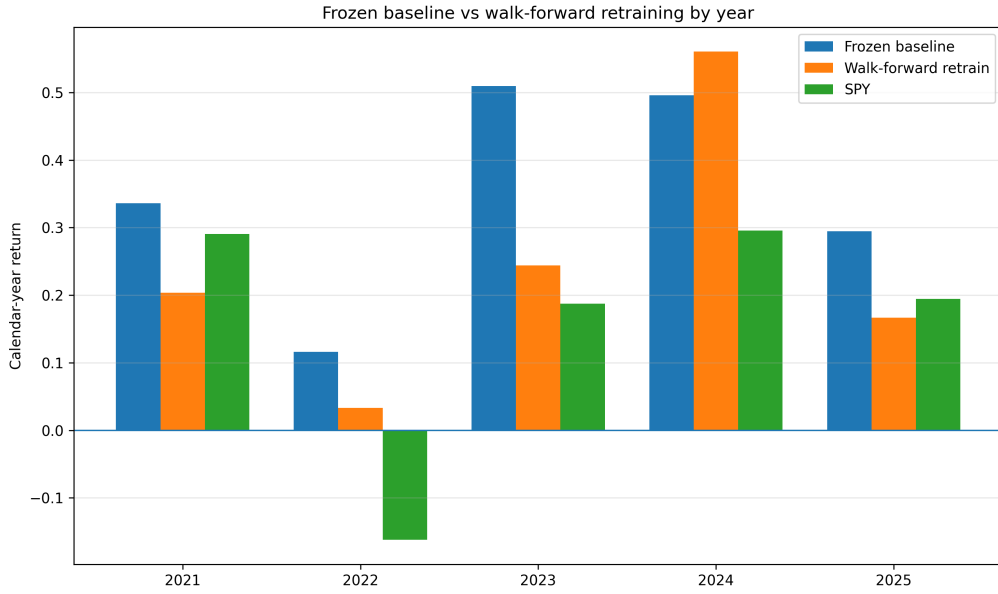


Figure 8: Year-by-year comparison of frozen baseline and walk-forward retraining

Table 23: Year-by-year walk-forward retraining diagnostic

Year	Frozen Top10	Walk-forward Top10	SPY	WF – Frozen	Top10 overlap
2021	33.60%	20.37%	29.06%	-11.13%	37.5%
2022	11.63%	3.31%	-16.22%	-6.68%	43.3%
2023	50.96%	24.43%	18.73%	-20.42%	36.7%
2024	49.57%	56.04%	29.55%	4.41%	37.5%
2025	29.45%	16.68%	19.44%	-10.31%	18.3%

Notes: Yearly returns are compounded from monthly rebalance returns. WF – Frozen is the annualized average monthly return difference between the walk-forward model and the frozen baseline. Top10 overlap is the average fraction of stocks selected by both specifications at the same monthly rebalance date.

Figure 8 and Table 23 show that the walk-forward underperformance is not driven by a single bad period. The walk-forward model underperforms the frozen baseline in 2021, 2022, 2023, and 2025, and outperforms it only in 2024. The largest annualized return gap occurs in 2023, not during the 2022 bear market. The walk-forward model still beats SPY in 2022, 2023,

and 2024, but it fails to match the stronger stock-selection performance of the frozen baseline.

The yearly diagnostic also shows that retraining changes the selected portfolio materially. The average Top10 overlap between the frozen and walk-forward specifications ranges from 18.3% to 43.3% depending on the year. This means that monthly retraining does not simply refine the frozen signal; it often changes the selected stocks substantially.

To check whether this instability comes from a major change in the variables used by the model, we also compare feature importances between the frozen baseline and the average walk-forward retrained models. Walk-forward feature importance is averaged across all 60 rebalance dates and seven random seeds, giving 420 fitted XGBoost models.

Table 24: Frozen versus walk-forward feature-importance stability

Comparison	Top20 overlap	Rank correlation	Importance correlation	Features
Frozen vs walk-forward	85.0%	0.9048	0.9048	68

Notes: Walk-forward feature importance is averaged across all monthly retrained XGBoost models and random seeds. Top20 overlap is the fraction of the twenty most important frozen-baseline features that also appear among the twenty most important walk-forward features. Rank correlation and importance correlation are Spearman correlations computed across the 68 common features.

Table 24 shows that the weaker walk-forward performance is not explained by a complete change in the model’s global feature structure. The top-20 feature overlap is 85.0%, and the Spearman correlation between frozen and walk-forward feature rankings is 0.9048. The most important walk-forward features include log market capitalization, gold returns, SPY returns and volatility, beta, price-to-book, aggregate bond returns, oil returns, momentum, VIX, and long-horizon relative strength. These are broadly consistent with the baseline interpretation that the model combines firm characteristics, momentum, risk, and cross-asset information.

The evidence therefore points to a more subtle explanation. Frequent retraining does not radically change which variables matter on average, but it changes the fitted nonlinear interactions and the resulting Top10 stock selections. Since monthly return ranks are noisy, the additional observations incorporated during the test period appear to increase portfolio-selection instability rather than improve adaptation to changing market conditions. The result supports retaining the frozen 2005–2018 model as the main specification, while clarifying that the baseline should be interpreted as a stable pre-test ranking model rather than as an adaptive online-learning system.

This walk-forward result is therefore not interpreted in isolation. Instead, it complements the broader robustness analysis by showing both the strength and the limits of the baseline specification. The main strategy remains economically strong across several robustness checks, but the paired tests show that additional complexity does not provide reliable improvement over the baseline. Frequent walk-forward retraining is significantly weaker than the frozen baseline, static minimum-variance optimization is also significantly weaker, and regime-aware forecasting does not produce statistically reliable improvement. Score-aware optimized portfolios are statistically indistinguishable from equal weighting, but they do not improve it.

4.9 Multiple-Testing Correction and Robustness Summary

Since the paper evaluates several specifications and reports many statistical tests, a multiple-testing correction is applied to assess whether the main significance claims survive adjustment for repeated testing. Table 25 applies the correction to the four pre-specified final baseline diagnostics. Table 26 summarizes the correction across broader families of robustness tests.

Table 25: Multiple-testing correction for final baseline diagnostics

Test	Raw p	BH-FDR q	Holm p	Bonferroni p	BH significant	Holm significant
Top-SPY	0.0050	0.0200	0.0200	0.0200	Yes	Yes
Top-Random	0.0112	0.0224	0.0336	0.0448	Yes	Yes
Top-Bottom	0.0407	0.0464	0.0814	0.1628	Yes	No
Spearman IC	0.0464	0.0464	0.0814	0.1856	Yes	No

Notes: The table applies multiple-testing corrections to the four pre-specified final baseline diagnostics. BH-FDR denotes the Benjamini-Hochberg false discovery rate correction. Holm and Bonferroni are more conservative family-wise error-rate corrections.

Table 26: Multiple-testing correction by test family

Test family	Tests	Raw $p < 0.05$	BH-FDR significant	Holm significant	Min. raw p	Min. BH-FDR q
Alternative models	24	9	0	0	0.0050	0.0672
Feature ablation	16	10	4	0	0.0040	0.0400
Horizon robustness	12	6	3	1	0.0028	0.0300
Hyperparameter robustness	8	3	0	0	0.0080	0.0640
Main baseline	4	4	4	2	0.0050	0.0200
Mega-cap removal	8	8	8	2	0.0014	0.0112
Multi-factor alpha	5	5	5	5	0.0024	0.0054
Paired regime tests	8	0	0	0	0.0837	0.1994
Paired alternative-model tests	20	7	0	0	0.0140	0.0700
Paired portfolio-optimization tests	8	2	0	0	0.0182	0.0728
Paired walk-forward tests	4	4	4	0	0.0154	0.0211
Point-in-time	4	4	4	4	0.0018	0.0072
Portfolio concentration	12	11	6	4	0.0012	0.0144
Portfolio optimization	10	4	1	1	0.0050	0.0500
Regime forecasting	8	3	0	0	0.0210	0.1155
Transaction costs	10	5	3	1	0.0050	0.0377
Walk-forward retraining	4	0	0	0	0.1375	0.4052

Notes: Corrections are applied within each test family. BH-FDR controls the false discovery rate within a family of related hypotheses. Holm is reported as a conservative family-wise error-rate correction. This table summarizes how many individual tests remain significant after correction.

Tables 25 and 26 lead to a more cautious but stronger interpretation of the statistical evidence. The four pre-specified baseline diagnostics remain significant after Benjamini-Hochberg false-discovery-rate correction. The Top-SPY and Top-Random tests also survive the more conservative Holm and Bonferroni corrections, while the Top-Bottom and Spearman IC diagnostics remain significant under BH-FDR but not under Holm or Bonferroni. Therefore, the strongest statistical evidence for the final baseline comes from its outperformance relative to SPY and random Top10 portfolios, with Top-Bottom and IC evidence providing supportive ranking validation. Among the paired tests, the walk-forward retraining underperformance remains significant after BH-FDR correction, whereas the paired alternative-model and portfolio-optimization differences should be interpreted as unadjusted paired-test evidence.

The family-level correction also clarifies the robustness results. Multi-factor alpha, point-in-time accounting, mega-cap removal, and several portfolio concentration tests remain significant after BH-FDR correction. By contrast, the regime-forecasting family and the alternative-model family do not retain significant tests after correction, which supports a cautious interpretation of those experiments. The regime-aware specifications are economically weaker than the baseline, and after multiple-testing correction they provide no statistically reliable evidence of improvement. This supports the paper’s main conclusion that HMM regimes are useful for interpreting conditional performance, but do not add robust incremental value when used directly in the forecasting model or portfolio-construction step.

Robustness Summary

Overall, the robustness tests support the main conclusion. First, the signal is robust to removing the largest mega-cap stocks. This addresses the possibility that performance is driven by a small number of dominant technology names. Second, alternative nonlinear tree models also perform well, while Ridge regression and the neural network are significantly weaker than XGBoost in paired tests. This suggests that the empirical result is not dependent on one exact tree-based algorithm, but that nonlinear tree-based methods are important. Third, portfolio concentration tests show the expected trade-off between concentration and diversification. Top5 generates higher return but lower stability, while Top20 improves Sharpe but reduces return. Top10 is a reasonable compromise.

Fourth, HMM regime experiments show that regimes are useful for conditional analysis but not for improving aggregate model performance. The paired regime tests provide no evidence that regime-aware forecasting improves the baseline. Fifth, portfolio optimization does not improve the equal-weight Top10 strategy. Static minimum variance is weaker than equal weighting in unadjusted paired tests, while score-aware and HMM score-aware optimization are statistically indistinguishable from equal weighting but do not improve it. Sixth, walk-forward retraining remains economically positive but is significantly weaker than the frozen baseline in paired tests. The yearly diagnostic shows that this underperformance is not driven by one isolated bad period, and the feature-importance comparison suggests that retraining changes portfolio selections more than it changes the global feature structure. Seventh, transaction costs do not eliminate the strategy’s economic value under realistic

cost assumptions.

Together, these tests indicate that the baseline finding is not fragile, although the multiple-testing correction shows that some marginal p-values around 0.03–0.05 should be interpreted cautiously. The strongest corrected evidence is for the main baseline outperformance relative to SPY and random Top10 portfolios, the multi-factor alpha robustness, the point-in-time accounting test, and the mega-cap removal test. More broadly, the robustness analysis shows that the main value comes from the stable XGBoost stock-ranking model itself: adding complexity through explicit regime conditioning, portfolio optimization, or frequent retraining does not provide reliable improvement over the frozen equal-weight Top10 baseline.

5 Discussion

5.1 Interpretation of Findings

The main empirical finding is that machine-learning stock ranking adds value, but explicit regime-aware forecasting and portfolio optimization do not. This has several implications.

First, cross-sectional ranking appears to be a more stable objective than direct return prediction. The model does not need to forecast exact returns accurately. It only needs to place stronger stocks above weaker stocks often enough for the Top10 portfolio to outperform.

Second, cross-asset market variables improve performance, whereas slow-moving macroeconomic variables degrade it. This suggests that market-implied information from traded assets is more useful for monthly stock selection than macroeconomic releases that update infrequently and may not help distinguish stocks cross-sectionally.

Third, HMM regimes matter for interpretation but not for improving the model. Regime 1 (which, in our case, corresponds to a bull-expansion state) is much more favorable to the baseline strategy than Regime 3 (which, in our case, corresponds to a crisis or extreme-stress state), but using regime labels directly in the forecasting model reduces aggregate performance. This implies that the global XGBoost model may already capture enough regime-related information through market and cross-asset features.

Fourth, equal weighting is difficult to beat. Although mean-variance optimization is theoretically appealing, the required inputs are noisy. Since the ML model already performs the main selection step, additional optimization may overreact to covariance and score estimation noise.

Finally, the transaction-cost analysis suggests that the strategy is not purely a frictionless backtest artifact. However, high turnover remains relevant. At 50 basis points per trade, the strategy loses statistical significance versus SPY.

5.2 Practical Implications

For practitioners, the results suggest that a relatively simple workflow can be effective: construct a robust feature set; train a nonlinear ranking model; select a concentrated set of high-ranked stocks; allocate equally; monitor regime-conditional performance; control turnover and implementation costs.

The results do not support replacing equal weighting with complex mean-variance optimization in this setting. The additional complexity does not improve performance and can reduce statistical significance.

The findings also suggest that regime models may be most useful as risk-monitoring and diagnostic tools. Rather than forcing separate forecasting models by regime, a practitioner may use regimes to understand when the model is likely to perform better or worse.

6 Limitations and Future Work

6.1 Limitations

Several limitations should be noted.

First, the universe construction may introduce survivorship bias since the stock list is based on current large-cap constituents rather than historical constituents.

Second, transaction costs are modeled as proportional costs based on turnover. The analysis does not include market impact, bid-ask spread variation by stock, tax effects, borrow constraints, or liquidity constraints.

Third, the test period contains 60 monthly observations. Although the cross-sectional panel is large, statistical tests on portfolio returns are limited by the number of rebalance periods.

Fourth, the strongest evidence is obtained for the frozen baseline model trained before the 2021–2025 test period. The walk-forward retraining experiment suggests that monthly re-estimation does not improve performance in this setting, so the baseline should be interpreted as a stable pre-test ranking model rather than as a fully adaptive online-learning system.

6.2 Future Research Directions

Future research could extend the paper in several ways.

First, the strategy could be tested internationally to determine whether the ranking signal generalizes beyond U.S. equities or to other securities.

Second, more realistic implementation assumptions could be added, including bid-ask spreads, market impact, liquidity filters, tax effects, and capacity constraints.

Third, the model could be extended to predict both ranking and uncertainty, enabling risk-aware stock selection without relying on noisy covariance optimization.

Fourth, regime detection could be improved by comparing HMMs with alternative regime models such as Markov-switching models, dynamic clustering, Bayesian state-space models, or macro-financial regime classifiers.

7 Conclusion

This paper studies whether market regime information improves machine-learning stock ranking and portfolio construction. The answer is nuanced. Regimes help explain performance, but they do not improve the main investment strategy.

The strongest result is the baseline XGBoost ranking model. A monthly equal-weight portfolio of the ten highest-ranked stocks earns a 34.24% CAGR, a Sharpe ratio of 1.279, and a statistically significant SPY-adjusted single-factor alpha and a positive multi-factor alpha statistically significant SPY-adjusted single-factor alpha and a positive multi-factor alpha after controlling for market, size, value, profitability, investment, and momentum factors during the 2021–2025 out-of-sample period. The model also produces significant Top-Bottom, Top-Random, Top-SPY, and Spearman information-coefficient results. These findings indicate that the model captures a meaningful cross-sectional return signal.

The regime-aware extensions do not strengthen this signal. HMM regimes are economically interpretable, and the strategy performs differently across them. However, adding regimes as predictive features, estimating separate models by regime, and using HMM-based covariance estimates in portfolio optimization all fail to improve aggregate performance. The evidence therefore does not support the hypothesis that explicit regime conditioning improves machine-learning stock ranking in this setting.

The portfolio construction results point in the same direction. More complex optimization methods do not outperform equal weighting within the selected Top10 stocks. The likely reason is that covariance and score-based optimization introduce additional estimation error into an already noisy forecasting problem. The simple equal-weight portfolio is more robust.

Moreover, the main statistical claims are checked using multiple-testing corrections. The core baseline diagnostics survive Benjamini-Hochberg false-discovery-rate correction, and the outperformance relative to SPY and random Top10 portfolios also survives more conservative Holm and Bonferroni corrections. Some marginal robustness results do not survive stricter corrections, so the broader robustness evidence should be interpreted cautiously. The central result, however, remains that the baseline XGBoost ranking model produces economically meaningful and statistically robust out-of-sample stock-selection performance.

Overall, the paper shows that machine learning can generate useful cross-sectional stock rankings, but it also shows the limits of adding complexity. The main economic value comes from the ranking model itself. Regime detection is useful as a diagnostic tool for understanding when the strategy performs well or poorly, but it adds little incremental value as a direct forecasting or portfolio-construction input.

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Data and Code Availability

Replication code is available at: https://github.com/carlospascualmiralles/carlospascualmiralles-RegimeAware_ML_PO. The repository contains the scripts used to construct features, estimate the expanding-window HMM regimes, train the machine-learning ranking models, form portfolios, compute robustness tests, and generate the tables and figures reported in the paper. The raw Bloomberg data cannot be redistributed because of data-licensing restrictions. The repository therefore provides the required input-file structure and variable names so that users with Bloomberg access can reproduce the analysis.

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A Technical Indicator Definitions

This appendix provides the definitions of the standard technical indicators used in the feature set. These variables are constructed using only information available at or before the prediction date.

Relative strength

Relative strength is measured by subtracting SPY returns from individual stock returns over matching horizons. This captures whether a stock has outperformed or underperformed the market over the same lookback window.

Let $R_{m,t}^{(-h)}$ denote the corresponding backward-looking return of the market benchmark, proxied by SPY. Relative strength is computed as

$$RS_{i,t}^{(h)} = R_{i,t}^{(-h)} - R_{m,t}^{(-h)}. \quad (54)$$

A positive value indicates that stock i has outperformed the market over the lookback window h .

Moving-average and trend indicators

The model includes price relative to 20-day, 50-day, and 200-day moving averages; the 20-day moving average relative to the 50-day moving average; and MACD-based trend indicators. These variables capture short-term and medium-term trend structure.

For stock i , let $P_{i,t}$ denote the adjusted closing price at date t . The simple moving average over a window of h trading days is defined as

$$SMA_{i,t}^{(h)} = \frac{1}{h} \sum_{\ell=0}^{h-1} P_{i,t-\ell}. \quad (55)$$

The price-to-moving-average variables are computed as

$$PriceSMA_{i,t}^{(h)} = \frac{P_{i,t}}{SMA_{i,t}^{(h)}} - 1, \quad h \in \{20, 50, 200\}. \quad (56)$$

The slope of the short-to-medium moving-average trend is proxied by

$$SMA20SMA50_{i,t} = \frac{SMA_{i,t}^{(20)}}{SMA_{i,t}^{(50)}} - 1. \quad (57)$$

A positive value of $PriceSMA_{i,t}^{(h)}$ indicates that the stock trades above its moving average over window h , while a positive value of $SMA20SMA50_{i,t}$ indicates that the short-term trend is above the medium-term trend.

The exponential moving average with span h is denoted by $EMA_{i,t}^{(h)}$. It is updated recursively as

$$EMA_{i,t}^{(h)} = \alpha_h P_{i,t} + (1 - \alpha_h) EMA_{i,t-1}^{(h)}, \quad \alpha_h = \frac{2}{h + 1}. \quad (58)$$

The MACD indicator is computed as the difference between the 12-day and 26-day exponential moving averages:

$$MACD_{i,t} = EMA_{i,t}^{(12)} - EMA_{i,t}^{(26)}. \quad (59)$$

The MACD signal line is the 9-day exponential moving average of the MACD series:

$$Signal_{i,t} = EMA_t^{(9)}(MACD_{i,t}). \quad (60)$$

Finally, the MACD histogram is defined as

$$Hist_{i,t} = MACD_{i,t} - Signal_{i,t}. \quad (61)$$

These indicators summarize whether the current price is above or below recent trend levels and whether shorter-term exponential trends are strengthening relative to longer-term trends.

Volatility and risk indicators

The model includes realized volatility over 5, 10, 20, and 60 days, average true range as a percentage of price, beta to SPY over 20 and 60 days, and correlation with SPY over 20 and 60 days. Realized volatility over a window of h trading days is computed as

$$\sigma_{i,t}^{(h)} = \sqrt{252} \text{sd} (r_{i,t-h+1}, \dots, r_{i,t}), \quad (62)$$

where $r_{i,t}$ denotes the one-day return of stock i .

The rolling market beta is estimated as

$$\beta_{i,t}^{(h)} = \frac{\text{Cov}(r_i, r_m)_{t,h}}{\text{Var}(r_m)_{t,h}}, \quad (63)$$

where r_m is the one-day SPY return and the covariance and variance are computed over the previous h trading days.

The rolling correlation with the market is

$$\rho_{i,m,t}^{(h)} = \frac{\text{Cov}(r_i, r_m)_{t,h}}{\sigma_{i,t}^{(h)} \sigma_{m,t}^{(h)}}. \quad (64)$$

Oscillators and volume indicators

The model includes RSI, stochastic %K, stochastic %D, Williams %R, one-day volume change, and volume relative to a 20-day moving average. These indicators are included to capture short-term overbought or oversold conditions, intraday range positioning, and abnormal trading activity.

Let $\Delta P_{i,t} = P_{i,t} - P_{i,t-1}$ denote the one-day price change. Define positive and negative price movements as

$$G_{i,t} = \max(\Delta P_{i,t}, 0), \quad L_{i,t} = \max(-\Delta P_{i,t}, 0). \quad (65)$$

The 14-day relative strength ratio is

$$RS_{i,t}^{(14)} = \frac{\frac{1}{14} \sum_{\ell=0}^{13} G_{i,t-\ell}}{\frac{1}{14} \sum_{\ell=0}^{13} L_{i,t-\ell}}. \quad (66)$$

The relative strength index is then

$$RSI_{i,t}^{(14)} = 100 - \frac{100}{1 + RS_{i,t}^{(14)}}. \quad (67)$$

The stochastic oscillator measures the position of the closing price within the recent high-low range. Let $H_{i,t}^{(14)}$ and $L_{i,t}^{(14)}$ denote the highest high and lowest low over the previous 14 trading days:

$$H_{i,t}^{(14)} = \max_{\ell=0, \dots, 13} High_{i,t-\ell}, \quad L_{i,t}^{(14)} = \min_{\ell=0, \dots, 13} Low_{i,t-\ell}. \quad (68)$$

The stochastic %K indicator is

$$\%K_{i,t} = 100 \frac{P_{i,t} - L_{i,t}^{(14)}}{H_{i,t}^{(14)} - L_{i,t}^{(14)}}. \quad (69)$$

The stochastic %D indicator is the 3-day moving average of %K:

$$\%D_{i,t} = \frac{1}{3} \sum_{\ell=0}^2 \%K_{i,t-\ell}. \quad (70)$$

Williams %R is computed as

$$W\%R_{i,t} = -100 \frac{H_{i,t}^{(14)} - P_{i,t}}{H_{i,t}^{(14)} - L_{i,t}^{(14)}}. \quad (71)$$

Volume dynamics are measured using both one-day volume growth and volume relative to its 20-day moving average. Let $V_{i,t}$ denote trading volume. The one-day volume change is

$$\Delta V_{i,t} = \frac{V_{i,t}}{V_{i,t-1}} - 1. \quad (72)$$

The volume-to-moving-average ratio is

$$VolRatio_{i,t}^{(20)} = \frac{V_{i,t}}{\frac{1}{20} \sum_{\ell=0}^{19} V_{i,t-\ell}}. \quad (73)$$

High RSI and high stochastic values indicate that the stock is trading near the upper end of its recent range. Williams %R is closer to zero when the stock is near the upper end of its range and closer to -100 when it is near the lower end. Volume variables capture whether the current price movement is accompanied by abnormal trading activity.